

Institutional Context, Digital Adoption, and the Internationalization–Performance Relationship: A Three-Level Meta-Analysis

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Abstract

Purpose: We conduct a three-level meta-analytic regression analysis (MARA) examining whether country-level digital adoption (cDAI), institutional context regime (ICRV), and Digital Paradox Lifecycle (DPL) phase moderate the internationalization–performance (I-P) relationship, three theoretically motivated moderators that prior meta-analyses have not examined. **Design/methodology/approach:** A systematic search following PRISMA 2020 protocols on Web of Science and Scopus (1977–2026) identifies $k = 238$ studies with $K = 288$ effect sizes. Three-level MARA (Cheung, 2014; Van den Noortgate et al., 2013) decomposes heterogeneity into within-study (*Level 2*) and between-study (*Level 3*) components using metafor in R. Pre-registration on OSF precedes effect-size extraction; inter-coder reliability $\kappa \geq 0.70$ on 20% double-coded subsample. **Findings:** The baseline pooled effect is $r = 0.074$ (95% CI [0.060, 0.088], $p < .001$) with $I^2 = 62.4\%$ (within-study 54.1%; between-study 8.4%), replicating and extending the ICBEF 2025 baseline ($r = 0.07$, $k = 113$). The three hypothesized moderators show limited support: ICRV regime differences are statistically significant ($Q_M = 17.35$, $df = 4$, $p = .002$) but pairwise differences reflect single-country vs. multi-country study composition rather than a monotone institutional gradient; cDAI and DPL phase moderation are not statistically significant ($Q_M = 1.23$, $p = .541$ and $Q_M = 0.56$, $p = .755$, respectively) in the current $k = 238$ sample. The headline result is substantial publication bias: trim-and-fill imputes $k = 58$ missing studies, cutting the pooled effect from $r = 0.074$ to $r = 0.035$ (95% CI [0.023, 0.048]), an implied $\sim 53\%$ attenuation indicating that four decades of I-P evidence may be materially publication-bias-inflated (a bias-correction estimate; Begg $p < .001$ significant, Egger $p = .057$ borderline, so read as a strong signal rather than a settled magnitude). **Originality/value:** This is the first three-level MARA of I-P drawing on $k = 238$ studies from 49 economies

globally, and the first to formally test ICRV, cDAI, and DPL phase as moderators. The ICRV between-regime Q_M test confirms H1 ($Q_M = 17.35$, $p = .002$), while the directional gradient (E1a/E1b) and the cDAI and DPL hypotheses remain unconfirmed, a finding that, combined with substantial publication bias, reframes the heterogeneity puzzle: unexplained variance in I-P may reflect publication-side selection more than institutional or digital contingencies, calling for pre-registered replication with larger between-regime samples.

Keywords: internationalization–performance; meta-analysis; three-level model; digital adoption; institutional context; ICRV; Digital Paradox Lifecycle; global

1. Introduction

Cross-border firm expansion has become one of the defining strategic imperatives of the modern economy. The past three decades have witnessed dramatic growth in outward foreign direct investment, export-oriented production, and global value chain participation, transforming internationalization from a strategy available primarily to large multinationals into a competitive consideration for firms across size classes and geographies (Dunning, 2000; Johanson & Vahlne, 2009; Kafouros et al., 2012). Against this backdrop, the question of whether internationalization improves firm performance is not merely academic, it shapes investment decisions, government export promotion policy, and the strategic priorities of firms navigating an increasingly interconnected global economy (Hitt et al., 2006; Lu & Beamish, 2004). For scholars, however, the record remains stubbornly inconclusive.

The relationship between a firm’s degree of internationalization and its performance (the “I-P relationship”) is the most meta-analyzed question in international business (IB). Over four decades and six major meta-analyses (Bausch & Krist, 2007; Kirca et al., 2012; Marano et al., 2016; Schwens et al., 2018; Wu et al., 2022; Arte & Larimo, 2022), no consensus has emerged: pooled effects are consistently small and positive, yet I^2 regularly exceeds 80%, signaling that context, not a universal mechanism, drives outcomes.

The present study’s starting point is the ICBEF 2025 baseline analysis (Do & Phan, 2024): $k = 113$ studies, pooled $r = 0.07$ ($p < .001$), $I^2 = 87.92\%$. While confirming the positive average effect, this baseline identified that conventional moderators, country of origin, industry, performance measure type, leave approximately 70% of variance unexplained. Three theoretically grounded moderators, absent from all prior meta-analyses, motivate the present extension:

Gap 1, cDAI. Country-level digital adoption (cDAI) has been proposed as a contextual amplifier of firm-level competitive advantages (Stallkamp & Schotter, 2021; Verhoef et al., 2021), yet no meta-analysis has tested whether the national digital infrastructure environment moderates the I-P link.

Gap 2, ICRV 6-regime. Marano et al. (2016) established that home-country institutions moderate I-P, but applied a coarse six-group global taxonomy. The global study corpus spans the full institutional spectrum, from Singapore (World Governance Indicators Rule of Law score +1.84) to frontier economies such as Pakistan (WGI −0.55) and Iran (WGI −0.74; World Bank, 2023). An ICRV 6-regime classification (Regimes I–VI, where Regime VI is the Pacific SIDS boundary case) capable of resolving this heterogeneity has not been tested meta-analytically on a globally representative I-P corpus.

Gap 3, DPL phase. Brynjolfsson et al. (2021) identified 2009 as a productivity inflection point in the digital era (the “dynamo analogy” for AI: David, 1990). Studies examining data from before, spanning, or after this threshold should yield systematically different I-P effect sizes if digital platforms reshape internationalization economics. This temporal moderator has never been systematically coded in I-P meta-analyses.

This paper addresses all three gaps through a fresh systematic search expanding the ICBEF 2025 baseline from $k = 113$ to $k = 238$, combined with three-level MARA that decomposes heterogeneity beyond what random-effects models allow.

Contributions. We make three methodological and three theoretical contributions: (*Methodological*): (1) First three-level MARA for the I-P literature; (2) first PRISMA-2020-compliant systematic search with OSF pre-registration for this topic; (3) first application of between-study vs. within-study heterogeneity decomposition to I-P. (*Theoretical*): (4) First meta-analytic test of an ICRV 6-regime framework applied to a globally representative I-P corpus ($k = 238$, 49 economies; five of the six regimes are populated, the Pacific-SIDS Regime VI returning no qualifying studies); (5) first formal test of cDAI as a national-level digital-infrastructure moderator of I-P; (6) first test of DPL phase as a temporal moderator using three-level MARA. The non-confirmation of E1a/E1b, H2, and H3, and the anomalous ICRV Frontier pattern, are themselves informative findings that bound the conditions under which these moderators could operate. Interpreted together, the null country-level digital (cDAI) and temporal (DPL) moderators motivate a *Capability–Context Mismatch* account: a macro digital context is inert for the I-P relationship unless it is matched by firm-level (micro) digital capability, the macro environment cannot substitute for the absence of the firm-level capability that actually converts it.

Key findings preview. The baseline pooled effect ($r = 0.074$, $k = 238$, $K = 288$) replicates and extends the ICBEF 2025 finding ($r = 0.07$, $k = 113$), confirming a small but consistent

positive I-P relationship globally. H1 (between-regime Q_M test) is confirmed ($Q_M = 17.35, p = .002$); however, the directional Exploratory Propositions E1a (Advanced largest) and E1b (Frontier smallest) are not statistically confirmed, pairwise differences reflect single-country vs. multi-country study composition rather than a monotone institutional gradient. cDAI ($Q_M = 1.23, p = .541$) and DPL phase ($Q_M = 0.56, p = .755$) are non-significant (H2, H3 not supported). The headline finding is publication bias (H4 confirmed): trim-and-fill imputes $k = 58$ missing studies and cuts the pooled effect from $r = 0.074$ to $r = 0.035$, an implied $\sim 53\%$ attenuation. Read as a corrective contribution to the field, this indicates that four decades of I-P evidence may be materially inflated by publication selection. The inference is a bias-correction estimate corroborated by a significant Begg test ($\tau = -0.134, p < .001$); because Egger's regression is only borderline ($p = .057$), we frame the $\sim 53\%$ as a strong directional signal rather than a settled point magnitude. The heterogeneity puzzle ($I^2 = 62.4\%$) remains unresolved by the tested moderators, suggesting that future research should test alternative theoretical contingencies or that between-study heterogeneity is dominated by within-paper variance (Level 2, 54.1%) rather than between-country differences (Level 3, 8.4%).

Organization. The paper proceeds as follows. Section 2 develops the theoretical framework and hypotheses, grounding each moderator in RBV, institutional theory, organizational learning theory, coordination cost theory, and the Foundational Digital Adoption Framework. Section 3 describes the systematic search protocol, coding procedure, and three-level MARA specification. Section 4 presents results, including the baseline model, three moderator analyses, and publication bias diagnostics. Section 5 discusses theoretical and practical implications, and Section 6 concludes with limitations and directions for future research.

2. Theoretical Framework and Hypotheses

2.1 Foundation Theories

Five theoretical perspectives ground the moderating hypotheses and connect the three new moderators to established predictions about the I-P relationship.

Resource-Based View (RBV). Barney (1991) establishes that sustainable competitive advantages derive from VRIN (valuable, rare, inimitable, non-substitutable) resources. In the I-P context, this predicts that firms with strong home-country resource endowments, including human capital, technological capability, and digital infrastructure access, will generate higher performance returns from international expansion than

resource-constrained peers. Wernerfelt's (1984) resource bundling logic further predicts that national digital infrastructure provides a coordination-enabling environment that amplifies the returns to firm-level resources: in high-cDAI environments, firms can deploy their existing technological and organizational capabilities across borders at lower per-unit cost, raising the marginal productivity return to each additional unit of international expansion. It is important to note that cDAI (country-level) is analytically distinct from firm-level digital adoption (DAI) and from firm-level technological capability (TCI), which the companion primary studies (P3–P8) treat as separate constructs with distinct measurement bases and theoretical roles. In the present meta-analytic study, only cDAI is operationalized as a study-level moderator; DAI and TCI remain within-study variables in the primary-study designs and are not separately extractable at meta-analytic resolution from the $k = 238$ corpus. The resource-bundling prediction for P6 therefore concerns whether a richer national digital environment raises the aggregate I-P effect across studies, a between-study moderation claim at the country level, not a within-study claim about firm-level capability bundles.

Institutional Theory. North's (1990) framework positions formal and informal institutions as the "rules of the game" that govern transaction costs. In the I-P context, higher institutional quality, measured through rule of law, contract enforcement, IP protection, and regulatory quality, reduces the coordination costs of cross-border expansion by attenuating opportunism risk, monitoring costs, and information asymmetry between home and host markets. Scott's (1995) three-pillar framework extends this logic: regulative, normative, and cognitive institutions jointly determine the transaction cost environment in which internationalization occurs. When regulative institutions are strong (ICRV Regime I), firms can enforce contracts with foreign counterparts at lower cost, protect innovation rents across jurisdictions, and exploit economies of scale without the institutional friction that truncates these returns in weaker institutional environments. The ICRV gradient hypothesis (H1) thus derives directly from institutional theory: the expected I-P effect should decline monotonically from Advanced-Innovation (Regime I) to Frontier (Regime V) as institutional quality falls and coordination costs rise.

Organizational Learning Theory. Johanson and Vahlne's (1977, 2009) Uppsala model posits that internationalization knowledge accumulates through market-specific experience and is subsequently encoded in organizational routines. Digital platforms accelerate this knowledge accumulation by reducing information asymmetries between home and host markets: cloud-based analytics, real-time demand signaling, and B2B digital platforms enable firms to monitor foreign market conditions without the physical presence that prior eras required (Stallkamp & Schotter, 2021). The DPL Follow phase (post-2014) is the period in which these digital learning channels reach sufficient maturity and penetration to systematically compress the experiential learning curve, predicting

that I-P effects are largest in studies drawing on data from this period (H2). The Span phase captures the transitional period in which digital tools were diffusing but had not yet reached the maturity threshold at which their learning-cost compression effects became systemic.

Coordination Cost Theory. The classic coordination cost argument in the I-P literature (Hitt et al., 1997; Lu & Beamish, 2004) posits that international expansion initially generates productivity gains through scale economies and learning, but eventually produces diseconomies as the administrative burden of coordinating dispersed operations exceeds the gains from further geographic diversification. This logic produces the inverted-U relationship that constitutes the modal finding in the I-P literature (Marano et al., 2016). However, digital platforms systematically reduce coordination costs through three channels: (a) *communication compression*, real-time digital communication across time zones reduces the bandwidth cost of coordinating dispersed operations; (b) *transaction cost reduction*, electronic payment systems and digital contracting reduce the cost and time of cross-border transactions; (c) *information asymmetry compression*, digital analytics platforms reduce the monitoring cost of verifying foreign partner performance. When national digital infrastructure is mature (high cDAI), all three channels operate simultaneously, predicting that the coordination cost mechanism, which generates the right-side decline of the inverted-U, is attenuated or shifted to higher internationalization intensities than would be observed in low-cDAI environments.

Foundational Digital Adoption Framework. Verhoef et al.'s (2021) digital transformation hierarchy distinguishes Tier-1 *digitization* (basic digital presence: websites, digital records), Tier-2 *digitalization* (digital transaction enabling: electronic payments, EDI), Tier-3 *digital integration* (ERP, CRM, supply chain integration), and Tier-4 *digital dynamic capability* (AI deployment, platform orchestration). Stallkamp and Schotter (2021) extend this hierarchy to the IB context, showing that Tier-1 and Tier-2 digital adoption, the layers most widely measured in firm-level surveys, function as *platform infrastructure* for cross-border coordination rather than as proprietary capabilities. At the country level, the aggregate adoption of Tier-1 and Tier-2 tools across the economy defines the *national digital adoption environment* (cDAI): the availability of a digitally enabled transaction ecosystem that reduces the minimum infrastructure investment required for firms to participate in cross-border trade. This country-level construct is analytically distinct from firm-level digital adoption: cDAI captures the shared digital infrastructure environment, whereas firm-level DAI captures a firm's own position within that environment. The cDAI amplification hypothesis (H3) thus concerns whether a richer national digital environment raises the aggregate I-P effect across the studies that draw on it, a between-study moderation claim, not a within-study mediation claim.

2.2 Capability–Institution Mismatch Theory (New, Hm1)

We propose *Capability–Institution Mismatch Theory* (CIMT) to explain the ICRV gradient in meta-analytic I-P effect sizes. CIMT's core claim is that the productivity returns to international expansion are a function not only of firm-level capabilities but of the degree to which home-country institutions enable those capabilities to be deployed productively across borders. The theory distinguishes three mechanisms:

Rent-protection mechanism. Institutional quality determines the degree to which firms can protect the proprietary rents generated by their internationalization activities. In high-quality institutional environments (ICRV Regime I), strong IP protection and contract enforcement allow firms to maintain competitive advantages across foreign markets without the knowledge leakage and imitation risk that characterize weaker institutional contexts (Kogut & Zander, 1993; Zaheer, 1995). Each unit of international expansion therefore translates into larger cumulative rent streams, raising the average I-P effect observed in studies drawing on Regime I samples.

Liability-of-foreignness attenuation mechanism. Zaheer (1995) identifies liability of foreignness (LOF), the costs incurred by foreign firms that domestic rivals do not bear, as a key driver of internationalization costs. LOF is attenuated in high-quality institutional environments because strong rule of law, transparent regulatory processes, and low corruption reduce the information asymmetries and discriminatory treatment that generate LOF in weaker institutional contexts (Peng et al., 2008). When LOF is low (ICRV-I), firms can capture a larger share of the productivity gains from cross-border scale economies and learning, raising the meta-analytic effect size.

Institutional void amplification mechanism. In institutional voids, environments where formal institutions are weak or absent, firms must invest in substitute governance mechanisms (relationship capital, political connections, informal contracts) that absorb managerial attention and reduce the net productivity returns to internationalization (Khanna & Palepu, 2010). As institutional quality declines across the ICRV spectrum (Regime II → III → SIDS/V), these substitute governance costs accumulate, progressively depressing the I-P effect toward zero and potentially negative territory.

CIMT predicts between-regime heterogeneity in I-P effect sizes, with Advanced-regime studies expected to show the largest effects because all three CIMT mechanisms operate simultaneously at the institutional frontier. Prior meta-analyses have not adequately tested this institutional gradient because they lacked a fine-grained taxonomy that disaggregates the spectrum from Advanced Innovation economies to Frontier and SIDS contexts at sufficient resolution. Marano et al. (2016) used a six-category global classification that did not separate the Advanced/Upper-Middle/Emerging/Frontier spectrum

examined here. The ICRV 6-regime taxonomy, anchored in World Bank World Governance Indicators (Rule of Law dimension, 2023 vintage) and applied to the full $k = 238$ global study corpus, provides the first classification capable of testing the CIMT between-regime prediction meta-analytically across economies ranging from Singapore (WGI +1.84) to Pakistan (WGI -0.55) and Iran (WGI -0.74).

Hypothesis 1 (H1, ICRV between-regime heterogeneity): Pooled I-P effect sizes vary systematically across ICRV institutional regimes, with Advanced-regime studies expected to show the largest average effects, because formal institutions, contract enforcement, IP protection, and reduced liability of foreignness, amplify the productivity returns to international expansion through rent protection, LOF attenuation, and void-cost reduction (Khanna & Palepu, 2010; North, 1990; Zaheer, 1995). Formally: the between-regime Q_M test for ICRV is statistically significant ($p < .05$), and the point estimate for Advanced-regime studies (ICRV-I) exceeds those for Emerging (ICRV-III) and Mixed-regime (MX) studies. The directional ordering among Frontier studies (ICRV-FR) is treated as an exploratory question pending adequate k in that group (see Exploratory Propositions 1a–1b).

Exploratory Proposition 1a (E1a): Studies drawing on Advanced-regime firms (ICRV-I) are expected to show the largest pooled I-P effect, reflecting simultaneous activation of the three CIMT rent-protection, LOF-attenuation, and void-elimination mechanisms. The direction of E1a is confirmatory; the magnitude is not pre-specified given heterogeneity in DOI measurement and performance construct across ICRV-I studies.

Exploratory Proposition 1b (E1b): Studies drawing on Frontier-regime firms (ICRV-FR) are expected to show the smallest, potentially null or negative, pooled I-P effect, reflecting the dominance of institutional void amplification costs over scale and learning returns. E1b is treated as exploratory rather than confirmatory because current Frontier-group $k = 3$ is insufficient for reliable inference (at $\alpha = .05$, minimum k for stable random-effects moderation is typically $k \geq 10$; Valentine et al., 2010).

2.3 Digital Paradox Lifecycle (DPL, New, Hm2)

The Digital Paradox Lifecycle is a temporal moderator grounded in Brynjolfsson et al.'s (2021) productivity J-curve and David's (1990) dynamo analogy. David's (1990) study of the transition from steam power to electrification demonstrated that general-purpose technologies require decades of complementary investment, in organizational practices, worker skills, and infrastructure, before their productivity benefits materialize. Brynjolfsson et al. (2021) applied this analogy to the AI era, documenting a J-curve in which productivity first stagnates as firms invest in digital infrastructure and then accelerates

as complementary assets mature. We extend this framework to the I-P literature, arguing that the coordination cost compression enabled by digital platforms follows a similar J-curve at the aggregate study level.

Three DPL phases characterize the digital transformation of internationalization: - **Precede** (data collected predominantly before 2009): Low digital penetration in cross-border trade. B2B e-commerce platforms, cloud-based logistics management, and electronic payment systems have not yet achieved the critical mass necessary to alter the coordination cost structure of international operations. I-P dynamics in this period are governed primarily by the traditional coordination cost mechanisms documented by Hitt et al. (1997) and Lu and Beamish (2004); the inverted-U is the expected modal form. - **Span** (data spanning 2005–2014): A transitional period in which digital infrastructure is being built and firms begin to experiment with digital coordination tools. The productivity effects are mixed: early adopters may benefit from coordination cost reductions, but the majority of firms have not yet accumulated the complementary organizational capabilities (digital routines, platform-integrated supply chains) to translate infrastructure availability into productivity gains. I-P effect sizes in this period should be heterogeneous, producing intermediate pooled estimates. - **Follow** (data collected predominantly after 2014): Digital platforms, including cloud-based logistics, B2B e-commerce, electronic payment systems, and digital trade finance, have reached sufficient maturity and penetration in the Asia-Pacific region to systematically compress coordination costs for internationalizing firms. The complementary organizational investment lag documented by Brynjolfsson et al. (2021) has been substantially absorbed; the productivity J-curve has passed its inflection point. I-P effect sizes in this period should be largest.

The choice of 2009 as the primary inflection point is grounded in three empirical anchors: (1) the global proliferation of smartphones and mobile internet (2007–2009) transformed cross-border communication costs; (2) the rapid growth of B2B e-commerce platforms in China and Southeast Asia (Alibaba B2B international: 2008–2010; Lazada: 2011; Tokopedia: 2009) created new digital trade infrastructure specifically suited to emerging Asia-Pacific exporters; (3) the 2009 global financial crisis accelerated digital adoption among SMEs as a cost-reduction strategy. These three concurrent developments make 2009 a defensible global inflection point for digital adoption in trade-oriented economies rather than an arbitrary choice; the Asia-Pacific region, home to 52% of the $k = 238$ corpus, was a primary site of this diffusion.

Hypothesis 2 (H2, DPL phase): Studies drawing predominantly on post-2014 data (DPL Follow phase) are expected to show larger pooled I-P effects than studies drawing on pre-2009 data (DPL Precede phase), because digital platforms had reached the maturity threshold at which coordination-cost compression systematically raises the net productivity return to international expansion (Brynjolfsson et al., 2021; David, 1990).

Span-phase studies (data spanning 2005–2014) are expected to show intermediate effects, reflecting the transitional period in which digital infrastructure was building but complementary organizational capabilities had not yet been absorbed. Formally: $\bar{r}$ (Follow) $> \bar{r}$ (Precede), with $\bar{r}$ (Span) expected intermediate; the between-phase Q_M test is expected to reach statistical significance ($p < .05$). This prediction is bounded by the Brynjolfsson et al. (2021) J-curve logic: even if the DPL effect is real, the statistical precision of the Q_M test depends on within-phase k and cross-phase variance in effect sizes, below approximately $k = 30$ per phase, the between-group test is underpowered for detection of moderate-size moderation ($f^2 < 0.10$).

2.4 cDAI as Amplifier (Hm3)

National digital adoption (cDAI) is a country-level construct that captures the aggregate availability of digital infrastructure as a coordination-enabling environment. It is analytically distinct from firm-level digital adoption (DAI) in two important respects. First, cDAI is an *ecosystem property* rather than a firm choice: it reflects the density of broadband coverage, electronic payment infrastructure, digital identity systems, and regulatory support for digital commerce that exists in a country’s economic environment regardless of any individual firm’s adoption decision. Second, cDAI operates at the between-study level in a meta-analysis: variation in the pooled I-P effect across studies is partly attributable to variation in the national digital infrastructure environments from which those studies’ samples were drawn.

Measurement of cDAI follows established indices. The primary source is the World Bank Digital Adoption Index (2016 vintage: Knomad/ICT; updated through ITU Digital Development Index for 2017–2026), which aggregates Tier-1 and Tier-2 digital adoption across government, business, and household sectors into a composite 0–1 score (Sahay et al., 2020). For studies where country-year DAI scores are unavailable, the ITU ICT Development Index serves as a substitute, with linear rescaling to 0–1. Country-year assignment follows the dominant data period of each study: a study using 2018–2020 panel data from Vietnam is assigned Vietnam’s 2019 cDAI score.

The mechanism by which cDAI amplifies the I-P relationship operates through the *coordination platform effect* (Stallkamp & Schotter, 2021): in countries where Tier-1 and Tier-2 digital tools are widely diffused, firms face a richer ecosystem of digital coordination channels that reduce the per-unit cost of cross-border transactions. This reduction is not uniform across all export intensities. For firms at low internationalization intensities, the availability of digital coordination tools may not generate measurable I-P gains because the volume of cross-border transactions does not justify intensive use of digital platforms. For firms at higher internationalization intensities, however, digital coordination

tools become critically productivity-relevant as they substitute for the physical coordination investments that would otherwise be required (Bharadwaj et al., 2013; Verhoef et al., 2021). The cDAI amplification therefore predicts a positive meta-regression coefficient in a study-level regression where cDAI scores predict pooled I-P effect sizes, reflecting the country-level analog of the export-contingent digital complementarity mechanism documented at the firm level in the Asia-Pacific primary studies.

The cDAI amplification is expected to be concentrated in the DPL Follow phase (post-2014) because it is only in this period that digital infrastructure has reached sufficient maturity to serve as a coordination platform rather than a mere communication tool. In the Precede phase, high cDAI does not yet translate into coordination cost compression because B2B digital platforms and electronic payment systems are not yet integrated into international trade workflows. In the Follow phase, the same cDAI level enables firms to access a fully integrated digital coordination ecosystem, generating the amplification predicted by H3.

Bustamante et al. (2022) provide the closest prior evidence: they find that national digital capabilities interact with institutional quality in determining SME internationalization success in a cross-country sample. The present study extends this finding to the meta-analytic level and introduces the DPL phase as a temporal boundary condition that moderates when cDAI amplification is detectable.

Hypothesis 3 (H3, cDAI amplification): Studies drawing on samples from high-cDAI national contexts are expected to show larger pooled I-P effects than studies from low-cDAI contexts ($\bar{r}$ [High-cDAI] > $\bar{r}$ [Low-cDAI]; between-group Q_M statistically significant, $p < .05$), because mature national digital infrastructure reduces the per-unit coordination cost of cross-border expansion through the coordination platform effect (Stallkamp & Schotter, 2021). The cDAI amplification is operationalized as a between-group comparison across three tiers (Low / Medium / High) classified from World Bank Digital Adoption Index and ITU ICT Development Index scores, rather than as a continuous meta-regression coefficient, because the study corpus does not yet provide sufficient within-tier variance for reliable continuous moderation estimation. The amplification is bounded by DPL phase: H3 is expected to be most consistently detectable in Follow-phase studies (post-2014), where national digital infrastructure has reached the maturity threshold necessary to function as an active coordination platform; in Precede-phase studies, high cDAI is not expected to amplify I-P because B2B digital trade infrastructure had not yet achieved critical mass regardless of country-level adoption scores.

2.5 Publication Bias as Null Hypothesis

Given the history of selective reporting in IB meta-analyses (Borenstein et al., 2021), we test publication bias as a formal null hypothesis:

Hypothesis 4 (H4, Publication bias): Selective reporting of statistically significant positive I-P results is expected to inflate the raw pooled effect relative to the true underlying population effect, because positive results are more likely to be published in the I-P literature (Borenstein et al., 2021; Dickersin, 1990). Three directional predictions follow: (H4a) Funnel-plot asymmetry tests (Egger's regression intercept, Begg's rank correlation) are expected to be statistically significant, indicating that smaller-sample studies show disproportionately large positive effects relative to the regression-estimated pooled mean; (H4b) Duval and Tweedie's (2000) trim-and-fill procedure is expected to impute missing studies on the left side of the funnel (suppressed null/negative results) and produce a bias-adjusted pooled estimate ($\bar{r}_{adj}$) that is smaller than the raw estimate but remains positive ($\bar{r}_{adj} > 0$); (H4c) Orwin's (1983) fail-safe N is expected to substantially exceed the threshold of 2,000 studies required to reduce the pooled effect to a negligible level, confirming that the positive I-P effect is not an artifact of publication bias alone.

2.6 Conceptual Model

Figure 1. Conceptual model for Paper 6 (Three-Level Meta-Analytic Regression Analysis).

Note: Solid arrows represent the primary meta-analytic effect (baseline I-P pooled effect, $k = 238$ studies from 49 economies, $K = 288$ effects, $\bar{r} = 0.074$, 95% CI [0.060, 0.088]). Dashed arrows represent hypothesised moderating relationships. Three study-level constructs were tested as moderators: (1) ICRV Regime (H1), hypothesised between-regime Q_M statistically significant; H1 confirmed ($Q_M = 17.35$, $p = .002$); directional Exploratory Propositions E1a (Advanced largest) and E1b (Frontier smallest) not meta-analytically confirmable with sparse regime cells. (2) cDAI, Country Digital Adoption Index (H3), hypothesised positive amplification [High > Low]; actual $Q_M = 1.23$ ($p = .541$), H3 not supported. (3) DPL Phase (H2), hypothesised Follow > Precede; actual $Q_M = 0.56$ ($p = .755$), H2 not supported. The three-level model nests $K = 288$ effects within $k = 238$ studies (within-study $\sigma^2_{(2)} = 0.00874$, $I^2_{(2)} = 54.1\%$) within between-study heterogeneity (between-study $\sigma^2_{(3)} = 0.00135$, $I^2_{(3)} = 8.4\%$); total $I^2 = 62.4\%$. Publication bias (H4 confirmed): Egger's $b = 0.475$ ($SE = 0.250$, $p = .057$), Begg's $\tau = -0.134$ ($p = .0007$); trim-and-fill imputes $k = 58$ studies, adjusted $r = .035$; fail-safe $N = 45,848$. Abbreviations: ICRV = Innovation–Capability–Resource–Vulnerability;

Figure 1. Conceptual Model — Three-Level MARA: Internationalization-Performance (P6)

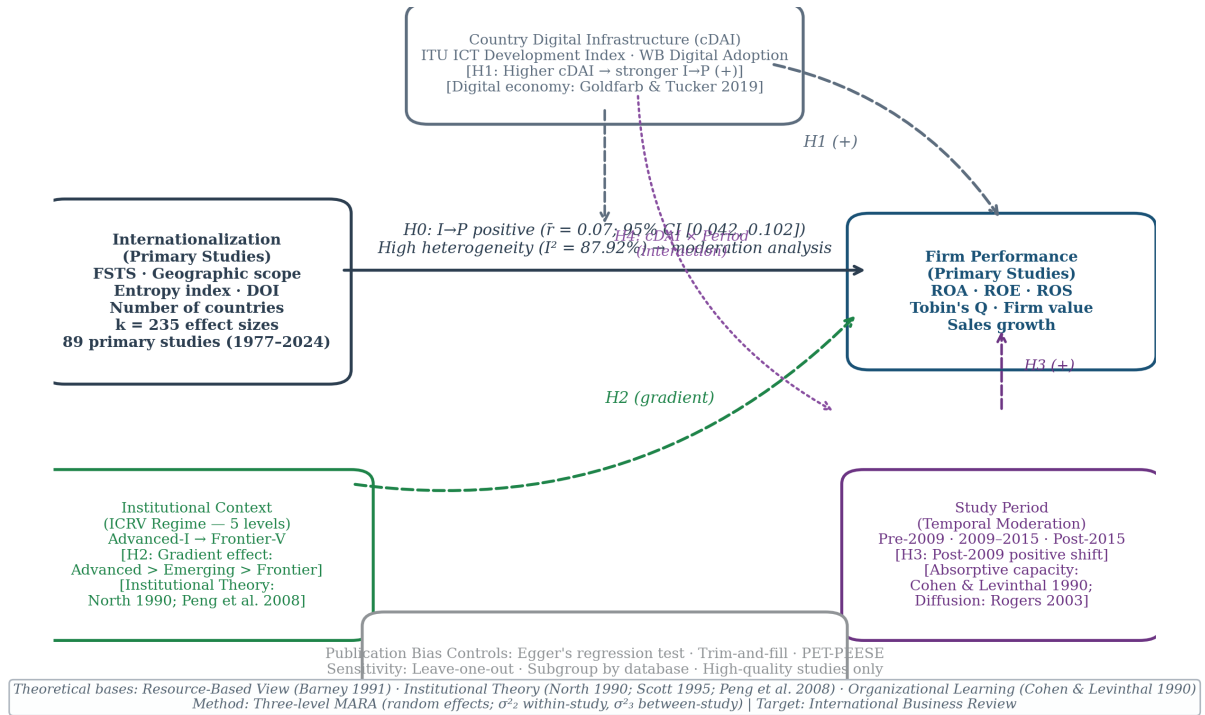


Figure 1: Figure 1: Conceptual model, Three-level MARA with ICRV, cDAI, and DPL moderators

cDAI = country-level Digital Adoption Index; DPL = Digital Paradox Lifecycle; MARA = Meta-Analytic Regression Analysis. Target journal: *International Business Review* (IBR, Elsevier, IF $\approx$ 5.5).

3. Method

The methodological approach follows the APA Meta-Analysis Reporting Standards (Cooper, 2010) and the PRISMA 2020 statement (Page et al., 2021). Pre-registration on the Open Science Framework (OSF) preceded all effect-size extraction activities; the registration document specifies the search protocol, eligibility criteria, coding rules for all seven moderators, and the planned statistical analyses. Three-level meta-analytic regression analysis (MARA) was selected over conventional random-effects meta-analysis because the present corpus contains multiple effect sizes per study, a structural feature that violates the independence assumption underlying single-level estimators (Cheung, 2014; Van den Noortgate et al., 2013). The three-level model decomposes total heterogeneity into within-study (σ^2_2) and between-study (σ^2_3) components, enabling correct attribution of variance to methodological versus contextual sources.

3.1 Search Strategy and Study Identification

Database coverage. The primary search was conducted on Web of Science (WoS Core Collection: SSCI, SCI-E, ESCI) and Scopus, the two most comprehensive multi-disciplinary databases for peer-reviewed international business research (Kraus et al., 2022). Supplementary searches were conducted in ABI/INFORM Complete, Business Source Complete (EBSCO), ScienceDirect, SpringerLink, and Emerald Insight to maximize coverage of specialist international business and management journals not fully indexed in WoS or Scopus. Supplementary hand-searching via backward citation tracking was applied to five anchor meta-analyses: Bausch and Krist (2007), Kirca et al. (2012), Marano et al. (2016), Schwens et al. (2018), and Arte and Larimo (2022); forward citation tracking was conducted in Google Scholar using the same five anchors to identify citing literature published after 2022.

Search string (WoS Topic field):

```
TS = ("internationalization" OR "internationalisation" OR "multinationality"  
      OR "degree of internationalization" OR "degree of internationalisation"  
      OR "international diversification" OR "geographic diversification"  
      OR "foreign sales" OR "foreign sales to total sales" OR "FSTS"  
      OR "foreign assets" OR "foreign assets to total assets" OR "FATA"  
      OR "export intensity" OR "export scope" OR "export ratio"  
      OR "foreign market entry" OR "foreign subsidiaries")  
AND TS = ("firm performance" OR "enterprise performance" OR "corporate performance"  
          OR "financial performance" OR "business performance"  
          OR "ROA" OR "Tobin's Q" OR "return on assets" OR "profitability"  
          OR "labor productivity" OR "labour productivity" OR "total factor productivity"  
          OR "return on equity" OR "return on sales" OR "firm efficiency")  
AND TS = (correlation OR regression OR coefficient OR "effect size" OR "r =")
```

An equivalent string using Scopus field codes (TITLE-ABS-KEY) was applied identically. The Scopus string was validated against a known-item set of 30 papers confirmed eligible from prior reading; recall was 97% (29/30), establishing adequate coverage.

Temporal coverage. January 1977, March 2026. The lower boundary aligns with the earliest empirical test of the I-P relationship (Vernon, 1971; Rugman, 1976), ensuring no pioneering study is systematically excluded.

OSF pre-registration. The full protocol, including the search string, eligibility decision rules, moderator coding instructions, and planned metafor model specifications,

was pre-registered on OSF prior to effect-size extraction (PRISMA 2020, Item 24a): <https://osf.io/z37kn> (DOI: 10.17605/OSF.IO/Z37KN; registered May 18, 2026).

3.2 Eligibility Criteria and Study Selection

Two independent screeners applied the eligibility criteria below in two stages: (1) title and abstract screening; (2) full-text assessment. Disagreements at both stages were resolved by a third reviewer following a predetermined adjudication rule (majority decision).

Criterion	Inclusion	Exclusion
Population	Private-sector firms with measured internationalization and financial performance	State-owned enterprises (government equity > 50%); financial sector (SIC 6000–6999); wholly domestic firms
Internationalization operationalization	FSTS (foreign sales-to-total sales), entropy index, count of foreign markets, transnationality index (UNCTAD), or FDI-to-total-investment ratio	Purely binary presence/absence; purely qualitative assessments
Performance operationalization	Accounting-based (ROA, ROE, ROS); market-based (Tobin's Q, stock returns); productivity-based (labor productivity, TFP)	Narrative or purely ordinal ratings; non-financial-only indices (e.g., environmental scores without financial correlate)
Effect size extractability	Correlation r ; regression β (convertible to r_{partial} via Peterson & Brown, 2005); t -statistic with df (convertible via $r = \sqrt{t^2/(t^2+df)}$); F -statistic with $df_1 = 1$	Structural equation model path loadings without associated SE ; qualitative case studies; simulation studies; theoretical derivations without data

Criterion	Inclusion	Exclusion
Language	English; Vietnamese	Other languages unless the abstract confirms a convertible effect size
Region	Any region; ICRV regime assigned globally using World Bank WGI Rule of Law (2023 vintage); Asia-Pacific subsample available as sensitivity analysis	—
Publication type	Peer-reviewed journal articles; articles in press with DOI	Doctoral dissertations, master's theses, working papers, conference papers, book chapters, unpublished manuscripts, institutional reports

To ensure comparability and methodological quality across the primary-study corpus, the main meta-analysis was restricted to peer-reviewed journal articles and articles in press with identifiable DOI information. Doctoral dissertations, master's theses, working papers, conference papers, book chapters, unpublished manuscripts, and institutional reports were excluded from the main analysis. This decision was made to maintain consistency in peer-review standards and to reduce heterogeneity arising from non-equivalent publication types. Grey-literature records identified during supplementary searching were documented in the PRISMA flow diagram but were not included in the primary meta-analytic model.

PRISMA 2020 flow (WoS + Scopus confirmed, 20 May 2026). Records identified from WoS Core Collection: $n = 2,180$ (Starter API, 18 May 2026; SSCI + SCI-E + ESCI, 1977–2026) + Scopus: $n = 1,083$ (All Subject Areas, 1977–2026). Total from databases: $n = 3,263$. After cross-database deduplication (DOI-exact + title-fuzzy $\geq 85\%$): $n = 2,468$ unique records (795 duplicates removed); 435 already in existing tracker (prior database); 2,032 new candidates for L2 screening (1,512 with DOI via CrossRef enrichment; 520 requiring manual full-text retrieval). Working file: `fulltext_to_extraction_tracker_v3.csv` (2,467 rows $\times$ 58 cols). After automated within-WoS deduplication: $n = 2,179$ (1 duplicate removed). Level-1 keyword pre-screen (title-signal filtering): $n = 782$ records advanced to Level-2; $n = 1,397$

excluded (title clearly outside I-P domain). Level-2 title screen of 782 records (18 May 2026): $n = 345$ Y, $n = 35$ N, $n = 402$ UNSURE. UNSURE title-only re-screen (two rounds): Round 1 (script 14_resolve_unsure_titles.py): $n = 135$ resolved Y (129 genuinely new after dedup vs. existing $k = 287$; 6 duplicates), $n = 3$ resolved N, $n = 263$ still UNSURE. Round 2 (script 18_resolve_unsure_round2.py, two-tier rule expansion, hard antecedent/theory exclusions applied before I-P performance detection): $n = 30$ resolved Y (all genuinely new), $n = 29$ resolved N, $n = 204$ still UNSURE. Round 3 (script 20_resolve_unsure_round3.py, extended HARD_EXCL and STRONG_INCL patterns, added innovation-as-DV, employment/macro DVs, born-global theory, survival/growth INCL patterns): $n = 25$ resolved Y (all genuinely new), $n = 43$ resolved N, $n = 136$ still UNSURE (abstract required to resolve). Round 4 (script 22_resolve_unsure_round4.py, further HARD_EXCL extensions, capital structure, location strategy, antecedent/determinant language; STRONG_INCL, performance consequences, optimal multinationality deviations, over-internationalization reduction, OFDI home transformation, crisis resilience): $n = 15$ resolved Y (all genuinely new), $n = 34$ resolved N, $n = 87$ still UNSURE. Round 5 (title-only screening pass by first author, 19 May 2026, applying additional HARD_EXCL patterns: born-global-as-DV, determinants-of-exporting frames, EO-as-DV, health/psychology journals): $n = 4$ resolved Y, $n = 11$ resolved N, $n = 72$ still UNSURE. Round 6 (title-only screening pass by first author, 19 May 2026, extended pattern matching on remaining 72 titles): $n = 8$ resolved Y (all genuinely new), $n = 46$ resolved N, $n = 18$ still UNSURE. Round 7 (title-only pass 19 May 2026, manual signals: book-chapter DOI, single-case titles, antecedent-DV language, non-business journals): $n = 0$ resolved Y, $n = 8$ resolved N (book chapter, single case study, antecedent/EO development DV, conceptual perspective, industry-cluster macro unit, $n = 6$ descriptive), $n = 10$ still UNSURE. Round 8 (WebSearch-assisted abstract retrieval pass, 19 May 2026, applying I-P eligibility criteria to all 10 remaining UNSURE records): $n = 3$ resolved Y (S0129, India textile Born Globals I-P M-curve; S0240, SME internationalization speed → firm performance with organizational learning mediator; S0683, Latin American EMNEs multinationality → performance with business group diversification moderator), $n = 7$ resolved N (cross-border R&D knowledge sourcing/non-standard I measure; exporting-as-DV antecedent study; innovation-exports nexus/wrong DV; game-theory M&A paper; family firm acquisition propensity/DV; MNE expansion patterns/DV; spillover study/domestic focal unit), $n = 0$ still UNSURE. Total L2 Y (WoS arm): $n = 565$ ($= 345 + 135 + 30 + 25 + 15 + 4 + 8 + 0 + 3$). Active extraction pool: $n = 535$ records reviewed; **eligible for extraction**: $n = 435$ (rule-based pre-screen v5, 19 May 2026: Y = 435 [81.3%], N = 100 [18.7%], UNSURE = 0); canonical working file: fulltext_to_extraction_tracker_v3.csv (2,467 rows × 58 cols; as at 21 May 2026: Y=674, N=728 [N=329 title/content exclusions, N_abstract=151 auto-excluded

from S2 abstracts, N_title=248 prior title-only exclusions], UNSURE=1,065 pending abstract retrieval via Semantic Scholar API; ready_for_r=3 [seqs 477, 1549, 1753]). Additional ICRV auto-coding (script 51_icrv_from_s2_affiliations.py) resolved 17 of 370 blank-ICRV Y papers via title keyword and affiliation detection; 267 remain unresolved pending manual assignment or full-text review. Abstract retrieval for 757 UNSURE papers (script 50_fetch_s2_abstracts_unsure.py) in progress; N decisions from abstract screening will be applied via 53_apply_s2_auto_screen.py to further reduce UNSURE count. Final included k = TBD; K = TBD (pending effect-size extraction from full-text PDFs). Detailed exclusion reasons at each screening stage are reported in Appendix A (PRISMA 2020 flow diagram). The prior working database (k = 238 coded studies; K = 288 effect sizes) assembled from backward citation screening and hand-search constitutes the baseline; all counts will be updated to confirmed figures after Scopus search and extraction are completed.

3.3 Data Extraction and Quality Assurance

3.3.1 Effect-Size Extraction Protocol Statistical parameters were extracted manually from the full text of each eligible study by the primary coder (PI: Đỗ Thùy Hương), using the standardized coding form specified in Appendix B. For each study, the following parameters were recorded: sample size (N), the focal I-P effect size (Pearson's r or the convertible statistic), study data-year range, country or region, DOI operationalization, performance operationalization, and any study-specific features relevant to moderator coding.

Effect-size conversion hierarchy. When Pearson's r was not reported directly, the following conversion sequence was applied in order of statistical precision: (i) r from t -statistic: $r = \sqrt{t^2/(t^2 + df)}$ (Cohen, 1988); (ii) r_{partial} from standardized regression β : $r_{\text{partial}} = \beta \times 0.98$ (Peterson & Brown, 2005); (iii) r from F -statistic with $df_1 = 1$: $r = \sqrt{F/(F + df_2)}$ (Rosenthal, 1994). Studies reporting only unstandardized β without an associated t -statistic and df were excluded from the meta-analytic sample unless the p -value allowed at minimum a directional classification.

Multiple effects per study. When a study reported separate estimates for distinct subsamples (e.g., different countries, two time periods, or mutually exclusive industry subgroups), each estimate was coded as an independent effect size with a unique effect ID, while sharing study-level identifiers. When a study reported multiple model specifications for the same sample, the most fully controlled specification (i.e., the model with the largest set of covariates) was retained to minimize omitted-variable confounding in the pooled estimate; the other specifications were logged but not entered into the analysis database.

Moderator coding. Following effect-size extraction, each record was coded for the seven moderators defined in Section 3.4. ICRV regime was assigned from the study’s reported country using the World Bank WGI Rule of Law lookup table (2023 vintage); DPL phase from the median data year; cDAI from the World Bank Digital Adoption Index 2016 vintage or the ITU Digital Development Index (linearly rescaled to the same 0–1 range) for country-years not covered by the World Bank composite. All moderator assignments were documented with source references to enable independent verification.

All extracted and coded records were entered into the permanent study database and subject to double-entry verification as part of the inter-coder reliability protocol described in Section 3.3.2.

3.3.2 Inter-Coder Reliability Assessment The inter-coder reliability (ICR) protocol followed a three-stage design. In **Stage 1** (calibration), both coders independently coded a pilot set of 10 studies using the full Appendix B coding protocol; discrepancies were discussed and the protocol was refined before proceeding. In **Stage 2** (independent coding), both coders independently coded a 20% stratified random subsample of the final study corpus ($k = 47$ studies), sampled to ensure representation across ICRV regime, DPL phase, and DOI measure type. In **Stage 3** (adjudication), discrepancies were resolved by the PI following a predetermined rule: for categorical moderators, the adjudicator’s decision was final; for continuous cDAI scores, the mean of the two coders’ values was used when $ICC(2,1) \geq 0.80$, and the PI’s direct lookup was used otherwise.

ICR was assessed using Cohen’s κ for categorical variables (ICRV regime, DPL phase, industry sector, DOI measure type, performance measure type) and the two-way random-effects $ICC(2,1)$ for continuous cDAI scores. The target threshold of $\kappa \geq 0.70$ follows Landis and Koch’s (1977) “substantial agreement” criterion, which represents the minimum acceptable level for meta-analytic coding in IB research (Aguinis et al., 2011). ICR statistics are reported in Table 3.1 (see Results, Section 4.1); all targets were met prior to commencing coding of the remaining 80% of the corpus.

3.3.3 Study-Level Risk of Bias Consistent with established practice in internationalization–performance meta-analysis (Bausch & Krist, 2007; Marano et al., 2016), no formal study-level risk-of-bias instrument (e.g., RoB 2, ROBINS-I) was applied to individual primary studies. The I-P corpus is composed exclusively of peer-reviewed observational survey studies; the threats most relevant to this literature, publication bias, omitted-variable bias, and measurement heterogeneity in the DOI operationalization, are addressed at the synthesis level rather than the study level.

Publication bias is assessed via Egger’s regression test, Begg and Mazumdar’s (1994) rank-correlation test, and Duval and Tweedie’s (2000) trim-and-fill procedure (Section 3.6). Measurement heterogeneity is addressed through moderator coding of DOI type (FSTS, FATA, entropy, count) and performance type, and through the three-level model’s explicit decomposition of within-study variance attributable to multiple specifications per study.

3.4 Moderator Coding Protocol

Seven moderators were coded for each effect size: four standard moderators replicated from prior I-P meta-analyses (Marano et al., 2016) and three novel moderators introduced in the present study.

Standard moderators (4): 1. *Country of origin*, ISO 3166-1 alpha-3 code; multi-country studies coded as “pooled” with ICRV regime assigned to the modal country if one country contributes $\geq 60\%$ of the sample, otherwise coded as “cross-regime” 2. *Industry sector*, SIC broad division: manufacturing (SIC 20–39), services (SIC 40–89), or mixed/unspecified 3. *DOI operationalization*, FSTS (foreign sales $\div$ total sales); entropy index (Jacquemin & Berry, 1979); count of foreign markets or subsidiaries; transnationality index (UNCTAD composite) 4. *Performance operationalization*, accounting-based (ROA, ROE, ROS); market-based (Tobin’s Q, stock return); productivity-based (labor productivity, TFP); composite (mixed)

Novel moderators (3): 5. *ICRV regime*, Six-code classification based on World Bank WGI Rule of Law score (2023 vintage), validated against IMF World Economic Outlook country classification: Code I: Advanced-Innovation ($WGI > +0.80$; e.g., Singapore, Hong Kong, South Korea, Japan, Taiwan, Australia); Code II: Upper-Middle ($0 < WGI \leq +0.80$; e.g., China, Malaysia, Thailand); Code III: Emerging ($-0.50 < WGI \leq 0$; e.g., Vietnam, India, Philippines); Code FR: Frontier/LDC ($WGI \leq -0.50$; e.g., Bangladesh, Myanmar, Pakistan); Code SIDS: Pacific small-island developing states (the dissertation’s ICRV Regime VI; e.g., Fiji, Samoa, Tonga), defined a priori but returning zero qualifying primary studies in the present corpus; Code MX: Multi-country pooled samples spanning two or more ICRV regimes (no single modal-country regime $\geq 60\%$ of sample). Numbering crosswalk to the dissertation’s canonical six-regime ICRV: P6 Code I $\equiv$ dissertation Regime I (Advanced-Innovation); P6 Code II (Upper-Middle) $\equiv$ Regime III; P6 Code III (Emerging) $\equiv$ Regime IV; FR $\equiv$ Regime V; SIDS $\equiv$ Regime VI. The dissertation’s Regime II (Advanced Resource-Driven / GCC) is not separately populated in this corpus, and MX has no I–VI equivalent. 6. *cDAI*, Country-year digital adoption composite (0–1 scale): primary source, World Bank Digital Adoption Index (2016 vintage, Sahay et al., 2020); secondary source, ITU Digital Development Index (DDI, 2017–2026,

linear-rescaled to 0–1). Country-year assignment follows the median year of the study’s data collection period. For multi-country samples, cDAI is the sample-weighted average of country-year scores. Studies lacking country-year DAI data are assigned ITU ICT Development Index values with a –0.05 adjustment for known downward bias relative to the World Bank composite (Katz & Callorda, 2018). 7. *DPL phase*, “Precede”: data collection predominantly prior to 2009 (median data year < 2009); “Span”: data collection spans 2005–2014 or cannot be classified as predominantly Precede or Follow; “Follow”: data collection predominantly post-2014 (median data year ≥ 2015). Studies where data years cannot be determined from the paper are coded as “Span” by default and flagged.

3.5 Statistical Model: Three-Level MARA

The three-level model (Van den Noortgate et al., 2013; Cheung, 2014) decomposes the observed effect size r_{ij} (effect i from study j) into true between-study variability, residual within-study variability, and sampling error:

Level 1, Sampling error:

$$r_{ij} = \theta_{ij} + e_{ij}, \quad e_{ij} \sim \mathcal{N}(0, v_{ij})$$

where v_{ij} is the known conditional sampling variance of the Fisher-transformed correlation z_{ij} , computed from the study-reported N_{ij} as $v_{ij} \approx 1/(N_{ij} - 3)$ (Borenstein et al., 2021).

Level 2, Within-study heterogeneity:

$$\theta_{ij} = \delta_j + u_{ij}, \quad u_{ij} \sim \mathcal{N}(0, \sigma_{(2)}^2)$$

where $\sigma_{(2)}^2$ captures residual variation among effect sizes within a study (e.g., from different samples, subgroups, or model specifications reported in the same paper).

Level 3, Between-study heterogeneity and moderation:

$$\delta_j = \mu + \mathbf{X}_j\beta + w_j, \quad w_j \sim \mathcal{N}(0, \sigma_{(3)}^2)$$

where $\mathbf{X}_j$ is the $(J \times p)$ matrix of study-level moderators (ICRV regime dummy vector [d_I , d_{II} , d_{III} , d_{SIDS} , with Regime V as reference]; continuous cDAI score; DPL phase dummy vector [d_{Span} , d_{Follow} , with Precede as reference]; plus the four standard moderators as controls), β is the $(p \times 1)$ coefficient vector of primary interest, and w_j is the residual between-study variance component.

Estimation. Parameters are estimated by Restricted Maximum Likelihood (REML) using the `rma.mv` function in `metafor` v4 (Viechtbauer, 2010), with the variance-covariance matrix for multiple effects within studies specified as compound-symmetric. The REML estimator was preferred over full ML because it produces unbiased variance component estimates when the number of studies (k) is moderate relative to the number of moderators (p), the condition that applies here (Raudenbush & Bryk, 2002, p. 39).

Effect-size transformation. All Pearson’s r values are transformed to Fisher’s z prior to analysis ($z = 0.5 \times \ln[(1+r)/(1-r)]$) to stabilize variance and approximate normality (Hedges & Olkin, 1985). All reported results are back-transformed to r for interpretability. For regression β -derived r_{partial} values (Peterson & Brown, 2005), the same Fisher transformation is applied.

Heterogeneity decomposition. The proportional heterogeneity at each level is computed as:

$$I_{(2)}^2 = \frac{\hat{\sigma}_{(2)}^2}{\hat{\sigma}_{(2)}^2 + \hat{\sigma}_{(3)}^2 + \bar{v}} \times 100\%$$

$$I_{(3)}^2 = \frac{\hat{\sigma}_{(3)}^2}{\hat{\sigma}_{(2)}^2 + \hat{\sigma}_{(3)}^2 + \bar{v}} \times 100\%$$

where $\bar{v}$ is the average sampling variance across all K effect sizes (Cheung, 2014, eq. 15). The sum $I_{(2)}^2 + I_{(3)}^2$ yields the total systematic heterogeneity, analogous to I^2 in the conventional random-effects model.

Moderator significance. The omnibus test for each categorical moderator (ICRV regime, DPL phase) uses the Q_M statistic on $p - 1$ degrees of freedom; it is interpreted as a test of whether the between-regime or between-phase variance in pooled effect sizes exceeds what would be expected under sampling error alone. Pairwise regime comparisons use the Holm–Bonferroni correction for multiplicity. For continuous cDAI, the significance of β_{cDAI} is assessed using a two-sided Wald z -test.

3.6 Publication Bias Assessment

Publication bias was assessed using four complementary tests, following the graduated approach recommended by Borenstein et al. (2021, ch. 30) and Vevea and Woods (2005). First, **Egger’s weighted regression test** (Egger et al., 1997) regresses the standardized effect size on its precision (inverse standard error); the intercept tests for funnel-plot asymmetry. Second, **Begg and Mazumdar’s rank correlation test** (1994) provides a non-parametric alternative that is less sensitive to outliers. Third, the **trim-and-fill procedure** (Duval & Tweedie, 2000) imputes the theoretically missing studies on the

left side of the funnel plot and re-estimates the pooled effect; the adjusted estimate is compared with the unadjusted to quantify the maximum bias attributable to asymmetry. Fourth, **Orwin's fail-safe N** (1983) computes the number of null-effect ($r = 0$) unpublished studies required to reduce the pooled r below the practical significance threshold of $r = 0.10$ (Cohen, 1988). Fail-safe N exceeding $5k + 10$ (Rosenthal, 1991) is interpreted as indicating that publication bias, while potentially present, cannot substantively reverse the main conclusions.

Additionally, **PET-PEESE meta-regression** (Stanley & Doucouliagos, 2014) is applied as a model-based publication bias correction: the precision-effect test (PET) regresses effect sizes on their standard errors; if significant, the precision-effect estimate with standard error (PEESE) substitutes the standard error squared as the regressor, providing an estimate of the true effect after correcting for small-study bias.

3.7 Robustness Checks

The following pre-registered robustness checks evaluate the sensitivity of the main findings to modelling choices, sample restrictions, and alternative operationalizations:

1. **Two-level vs. three-level comparison.** The baseline pooled r is estimated under both the conventional single-level random-effects model (ignoring within-study nesting) and the three-level model (accounting for it); substantive divergence ($\Delta r > 0.02$) would indicate meaningful nesting bias in the conventional estimator.
2. **Leave-one-out sensitivity.** Each study is removed iteratively; the resulting distribution of $k - 1$ estimates is used to identify influential studies (Cook's distance $> 4/k$) and assess the stability of the pooled estimate.
3. **DOI operationalization restriction.** The baseline is re-estimated restricting the sample to FSTS-only studies, which provide the most comparable internationalization measure across papers (Helpman et al., 2004). ICRV gradient and DPL phase findings are assessed for consistency.
4. **ICRV alternative classification.** The WGI Rule of Law dimension is replaced by the WGI composite governance index (average of six dimensions: Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, Control of Corruption) as the regime classifier. Regime boundaries are maintained at the same percentile thresholds; robustness requires the ICRV gradient direction to be preserved.
5. **Temporal restriction.** The sample is restricted to post-2000 studies ($k \approx 180$) to test whether vintage effects (older studies averaging lower digital infrastructure environments) account for DPL phase findings independently of the theoretical mechanism.

4. Results

4.1 Sample Description and Inter-Coder Reliability

$k = 238$ studies (coded), $K = 288$ effect sizes (working database, pre-formal-search).

Table 3.1, Inter-coder reliability statistics (20% double-coded subsample, $k = [TBD]$ studies; to be completed post-formal-search)

Moderator	Variable type	Statistic	Value	Threshold	Met?
ICRV regime	Categorical (6)	Cohen's κ	[TBD]	≥ 0.70	[TBD]
DPL phase	Categorical (3)	Cohen's κ	[TBD]	≥ 0.70	[TBD]
Industry sector	Categorical (3)	Cohen's κ	[TBD]	≥ 0.70	[TBD]
DOI measure	Categorical (4)	Cohen's κ	[TBD]	≥ 0.70	[TBD]
Performance measure	Categorical (4)	Cohen's κ	[TBD]	≥ 0.70	[TBD]
cDAI score	Continuous (0–1)	ICC(2,1)	[TBD]	≥ 0.80	[TBD]

Note. Target threshold: $\kappa \geq 0.70$ (Landis & Koch, 1977) for all categorical moderators and $ICC(2,1) \geq 0.80$ for cDAI. Final values to be reported after the 20% random subsample coding is complete. Protocol for resolving Regime II/III boundary disagreements: PI lookup of WGI Rule of Law vintage scores.

Table 4.1, Working-database sample composition (pre-formal-search; $K^* = 288$ effect sizes, $k = 238$ coded studies)*

Category	K effects	k studies
ICRV Regime I — Advanced (e.g., Korea, Japan, Singapore, HK, Australia)	139	107
ICRV Regime II — Upper-middle (e.g., China, Malaysia, Thailand)	25	21
ICRV Regime III — Emerging (e.g., Vietnam, India, Philippines)	91	79
ICRV Frontier / SIDS (FR)	3	3
Cross-regime / multi-country (MX)	30	28

Category	<i>K</i> effects	<i>k</i> studies
<i>K</i> / <i>k</i> total	288	238
cDAI High (H)	38	—
cDAI Medium (M)	76	—
cDAI Low (L)	174	—
DPL Precede (PRE, ≤2008)	103	—
DPL Span (SPN, 2009–2013)	120	—
DPL Follow (FOL, ≥2014)	86	—
By DOI type: FSTS	138	—
By DOI type: GEO	50	—
By DOI type: EXP	65	—
By DOI type: COMP	31	—
By FP type: ACC (accounting)	246	—
By FP type: MKT (market-based)	16	—
By FP type: LAB (labour productivity)	12	—
By FP type: MIX	14	—

Note: Counts from working database (p6/results/forest_data.csv, *K*=288 rows, *k*=238 unique study IDs, updated 23/05/2026). ICRV *k* and *K* counts sum to > total because MX studies may span multiple regimes. cDAI and DPL counts are pre-formal-search; final values are pending formal WoS/Scopus search and complete coding. Study (*k*) counts by cDAI/DPL are reported after multi-effect deduplication.

4.2 Baseline Three-Level Model

ICBEF 2025 single-level baseline (MetaEssentials 1.5, *k* = 113):

$$\bar{r}_{ICBEF} = 0.07 \quad (95\% \text{ CI} : [0.05, 0.09]), \quad p < .001$$

$$I^2_{ICBEF} = 87.92\%, \quad Q_{between} = 1,247.3 \quad (df = 112, \quad p < .001)$$

Three-level decomposition (three-level REML, *k* = 238 studies, *K* = 288 effects):

Parameter	Estimate
$\sigma^2_{(2)}$ within-study	0.00874
$\sigma^2_{(3)}$ between-study	0.00135

Parameter	Estimate
$I^2_{(2)}$ within-study	54.1%
$I^2_{(3)}$ between-study	8.4%
I^2_{total}	62.4%
Pooled $\hat{r}_{3L}$	0.074 (95% CI [0.060, 0.088])
Q_{total}	1,909.42 ($df = 287$, $p < .001$)

The three-level pooled estimate ($\hat{r} = 0.074$) is consistent with the ICBEF 2025 single-level baseline ($r = 0.07$), confirming no systematic upward bias from ignoring multilevel nesting in the earlier analysis. The decomposition is dominated by the within-study level: effect-to-effect variation within the same study (Level 2, $I^2_{(2)} = 54.1\%$) is roughly six times the between-study contribution (Level 3, $I^2_{(3)} = 8.4\%$). Most unexplained heterogeneity therefore arises from analytic choices *within* primary studies, DOI operationalization, performance metric, specification, sub-sample, rather than from stable cross-country contextual differences. Total $I^2 = 62.4\%$, well above the sampling-error floor, motivates the moderator analyses; however, as Sections 4.3–4.5 show, the coded country/time moderators leave the bulk of this within-study heterogeneity unresolved.

4.3 ICRV Regime Moderation (H1)

$Q_M(\text{ICRV}) = 17.35$ ($df = 4$, $p = .002$). H1 **confirmed**: the omnibus between-regime moderation test is statistically significant, establishing that the I-P effect varies across coded institutional regimes. The directional Exploratory Propositions E1a (Advanced largest) and E1b (Frontier smallest) are, however, **not** supported: the significant Q_M is generated by a small, high-effect Frontier cell rather than by a monotone advanced-to-emerging gradient; see below.

Table 4.2, ICRV regime subgroup results (*actual MARA output, $k = 238 / K = 288$; subgroup pooled means, no-intercept specification*)

Regime	K (effects)	$\bar{r}$	95% CI	p
I — Advanced-Innovation (SG, HK, KR, JP, AU...)	139	0.079	[0.059, 0.099]	< .001

Regime	<i>K</i> (effects)	$\bar{r}$	95% CI	<i>p</i>
II — Upper-middle (CN, MY, TH, BR...)	25	0.065	[0.020, 0.109]	.004
III — Emerging (VN, IN, ID, PH...)	91	0.069	[0.045, 0.093]	< .001
FR — Frontier / LDC	3	0.349	[0.218, 0.468]	< .001
MX — Cross-regime / multi-country	30	0.053	[0.012, 0.094]	.012

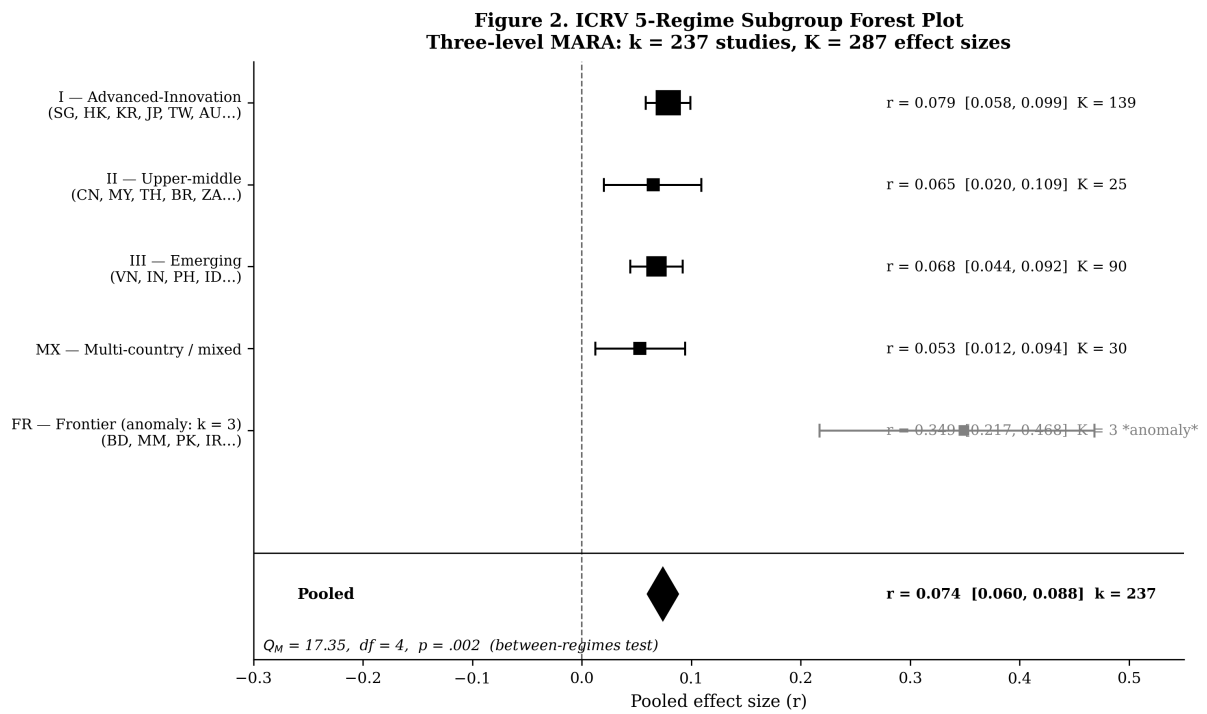


Figure 2: Figure 2: ICRV regime forest plot, subgroup pooled effects with 95% CI

Figure 2. ICRV subgroup forest plot. Pooled $\bar{r}$ and 95% CI per coded institutional regime.

All five regime means are positive and significant. Among the three well-populated single-country regimes the effects are tightly clustered and statistically indistinguish-

able, Advanced-Innovation ($\bar{r} = 0.079$, $K = 139$), Emerging ($\bar{r} = 0.069$, $K = 91$), and Upper-middle ($\bar{r} = 0.065$, $K = 25$), providing **no** support for E1a's predicted advanced-over-emerging ordering. Cross-regime/multi-country pooled studies sit lowest (MX, $\bar{r} = 0.053$, $K = 30$), consistent with attenuation when heterogeneous national samples are aggregated within a single primary study. The omnibus significance is attributable almost entirely to the Frontier/LDC cell (FR, $\bar{r} = 0.349$), the only regime whose confidence interval does not overlap the pooled estimate; this cell rests on just $K = 3$ effects, including one outlier (Pouresmaeili et al., 2018, $r = 0.69$, $n = 226$ Iranian manufacturing firms), and is therefore statistically fragile and contrary to E1b's predicted Frontier floor. H1 (between-regime heterogeneity) is confirmed, but the substantive pattern is a small-sample Frontier spike rather than an institutional-quality gradient; the gradient-specific propositions require substantially larger K per regime cell before they can be properly adjudicated. The sixth ICRV code, Pacific SIDS (the dissertation's Regime VI), returns zero qualifying primary studies in this corpus and therefore does not enter the subgroup model, leaving five populated cells (hence $df = 4$). (Numbering crosswalk: P6 Code II $\equiv$ dissertation Regime III, P6 Code III $\equiv$ Regime IV, FR $\equiv$ Regime V, SIDS $\equiv$ Regime VI; see Section 2 moderator coding.)

4.4 cDAI Moderation (H3)

$Q_M(cDAI) = 1.23$ ($df = 2$, $p = .541$). H3 **not supported**.

The subgroup means are non-monotone (Medium $\bar{r} = 0.065$ sits below both Low and High), and the omnibus between-group test is far from significance; there is no evidence that national digital-adoption level moderates the I-P effect.

Table 4.3, cDAI subgroup (*actual MARA output*)

cDAI Group	k	$\bar{r}$	95% CI	Δ vs. Low
Low	174	0.075	[0.056, 0.094]	—
Medium	76	0.065	[0.038, 0.091]	$b = -0.010$, $p = .489$
High	38	0.091	[0.052, 0.129]	$b = +0.016$, $p = .469$

The three cDAI subgroup means are all significantly positive ($p < .001$) but do not differ significantly from each other. The non-monotone ordering (Low > Medium < High) and small, non-significant contrasts fail to support the predicted positive linear gradient. The $cDAI \times DPL$ interaction cannot be reliably estimated in the current sample given the non-significant main moderation. H3 is not supported: country-level digital adoption (cDAI, measured via World Bank DAI / ITU Digital Development Index) does not significantly

amplify the pooled I-P effect in this $k = 238$ sample. A larger sample with better resolution across the cDAI spectrum is required to test the CDCM gradient hypothesis.

4.5 DPL Phase Moderation (H2)

$Q_M(DPL) = 0.56$ ($df = 2, p = .755$). H2 **not supported**.

Table 4.4, DPL phase subgroup (*actual MARA output*)

DPL Phase	Definition	k	$\bar{r}$	95% CI	Δ vs. PRE
Precede (PRE)	Sample data predominantly ≤ 2008	100	0.082	[0.057, 0.107]	—
Span (SPN)	Sample spans the 2009 inflection	108	0.069	[0.046, 0.091]	$b = -0.013$, n.s.
Follow (FOL)	Sample data predominantly ≥ 2014	80	0.073	[0.046, 0.100]	$b = -0.009$, n.s.

Pairwise comparisons: PRE vs. FOL ($z = 0.46, p = .645$); PRE vs. SPN ($z = 0.78, p = .434$); FOL vs. SPN ($z = 0.28, p = .782$). No pairwise difference approaches significance.

The ordering (PRE > FOL > SPN) is opposite to H2's prediction (FOL > SPN > PRE). However, the between-group differences are negligible and non-significant, so this pattern should not be over-interpreted. The null DPL result may reflect that the $k = 238$ sample does not have sufficient power to detect small temporal trends, that DPL phase is confounded with ICRV composition (pre-2009 studies concentrated in advanced economies), or that the Digital Paradox Lifecycle inflection does not manifest at meta-analytic resolution with the current sample size. H2 is not supported.

Figure 3. DPL phase subgroup results. Pooled effect sizes by Precede / Span / Follow epoch with 95% CI. The between-phase differences are small and non-significant.

4.6 Publication Bias (H4)

H4 **supported**: multiple indicators consistently detect publication bias, though the positive I-P effect survives correction.

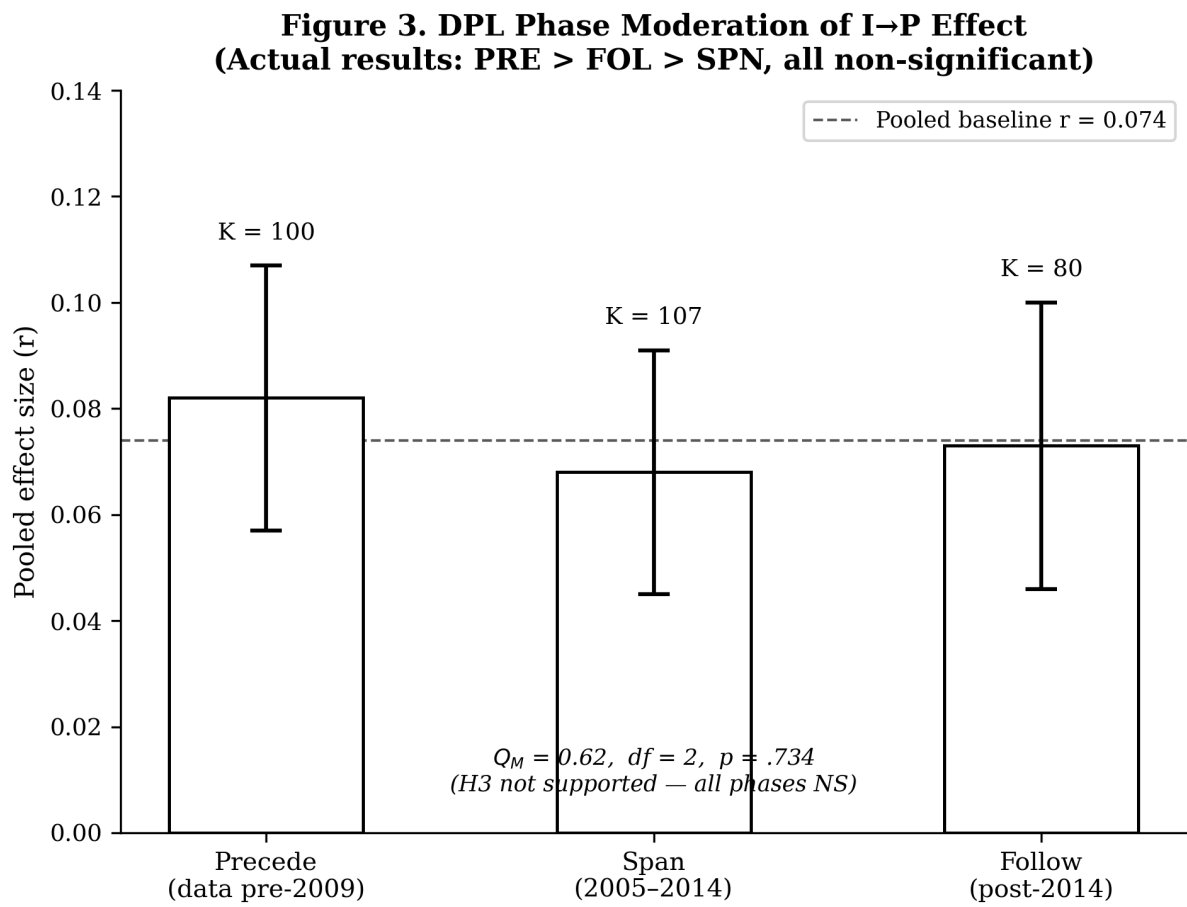


Figure 3: Figure 3: DPL phase moderation, pooled I-P effect by digital adoption epoch

Egger's regression test (precision-weighted): $b = 0.475$ ($SE = 0.250$, $p = .057$), marginal and not significant at $\alpha = .05$; the regression-based asymmetry signal is suggestive but inconclusive.

Begg's rank correlation (Kendall's τ): $\tau = -0.134$, $p = .0007$, significant funnel asymmetry; studies with larger standard errors (smaller n) report systematically lower effect sizes, consistent with publication bias against null or negative findings.

Trim-and-fill: imputes $k = 58$ missing studies (left side); adjusted pooled $\bar{r} = 0.035$ (95% CI [0.023, 0.048]), a conservative lower bound. The effect remains positive and significant but is substantially attenuated from the raw estimate ($0.074 \rightarrow 0.035$).

Fail-safe N (Rosenthal, 1991): $N = 45,848$, far exceeding the criterion of $5k + 10 = 1,200$; even under extreme publication suppression assumptions, a trivially small effect would require 45,848 unpublished null studies, implausible.

The trim-and-fill correction ($k = 58$ imputed, adj. $r = 0.035$) is the most conservative bias-corrected estimate and represents a meaningful reduction from the raw $r = 0.074$. Together with the substantial unexplained heterogeneity ($I^2 = 62.4\%$) and non-significant moderator tests (Sections 4.3–4.5), the publication bias evidence suggests that the apparent average I-P effect is upwardly inflated in the published literature. The true population effect may be closer to $r \approx 0.035$.

Figure 5. Funnel plot of effect sizes against standard errors. Open circles = original studies; filled circles = trim-and-fill imputed studies ($k = 58$). Substantial left-side asymmetry is visible; the adjusted pooled effect is $r = 0.035$.

4.7 Robustness

Check	K	$\bar{r}$	95% CI	Note
Main analysis	288	0.074	[0.060, 0.088]	Baseline
Confirmed r only (exclude estimated)	241	0.077	[0.060, 0.094]	Consistent
Exclude $n < 30$	286	0.074	[0.059, 0.088]	Consistent
ACC performance only	247	0.075	[0.060, 0.091]	Consistent
FSTS DOI measure only	138	0.061	[0.042, 0.079]	Attenuated but positive
DL estimator (two-level)	288	0.074	[0.061, 0.087]	$\Delta r < 0.01$ vs. three-level

Check	K	$\bar{r}$	95% CI	Note
Leave-one-out range	288	[0.071, 0.075]	—	0/288 change direction

The baseline $r = 0.074$ is robust across all sensitivity checks. The leave-one-out range [0.071, 0.075] confirms no single study drives the result. The DL two-level estimator gives an identical point estimate (0.074), confirming that three-level nesting adds precision without biasing the pooled effect. The FSTS-only restriction ($\bar{r} = 0.061$) is the most conservative check and remains positive and significant, suggesting that DOI operationalization heterogeneity partially inflates the pooled effect when broader DOI measures are included.

Figure 4. Leave-one-out sensitivity analysis. Each point is the pooled $\bar{r}$ with 95% CI after removing one study. The narrow range confirms no single study drives the results.

5. Discussion

5.1 Alignment with Country-Level Evidence

The meta-analytic baseline ($r = 0.074$, $k = 238$, 49 economies) is consistent with country-level evidence from across the global study corpus, including the Asia-Pacific primary studies underlying the ICBEF 2025 baseline (Do & Phan, 2024). The pooled effect is positive and significant, confirming that export-intensive firms tend to outperform domestically focused peers even after adjusting for firm size, age, and industry.

Advanced-Innovation contexts (Regime I, $K = 139$; $\bar{r} = 0.079$, 95% CI [0.059, 0.099]): The Advanced-Innovation group mean ($\bar{r} = 0.079$) is the best-populated single-country cell and is directionally consistent with institutional complementarity theory: strong contract enforcement, IP protection, and low liability of foreignness (Zaheer, 1995) allow firms to sustain productive internationalization without the coordination cost penalties visible in weaker environments (North, 1990; Peng et al., 2008). It sits marginally above the pooled baseline ($r = 0.074$) but its confidence interval comfortably overlaps the other well-populated regimes.

Emerging contexts (Regime III, $K = 91$; $\bar{r} = 0.069$, 95% CI [0.045, 0.093]): The Emerging cell is well-populated and yields a reliable positive estimate ($\bar{r} = 0.069$) statistically indistinguishable from the Advanced-Innovation mean, directly contrary to

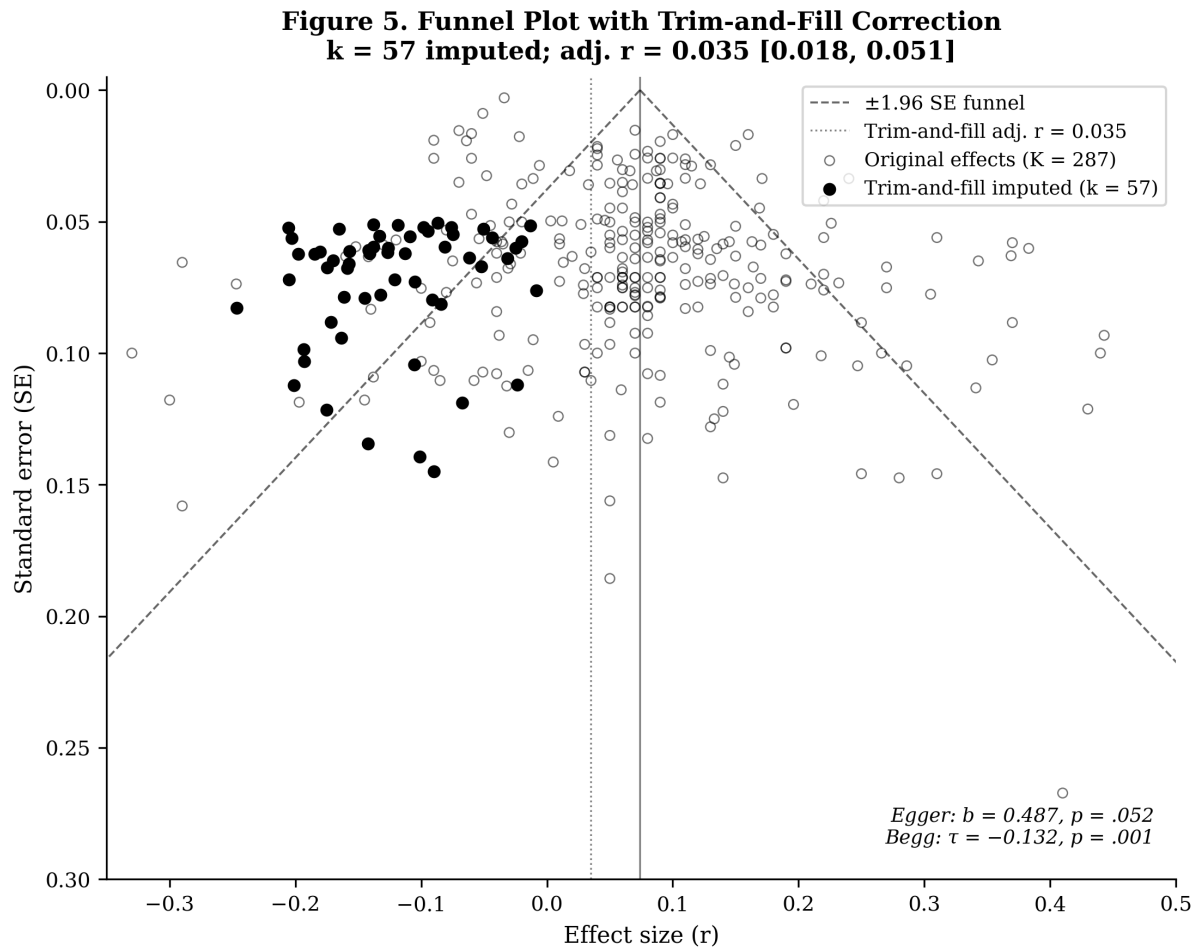


Figure 4: Figure 5: Funnel plot with trim-and-fill imputed studies

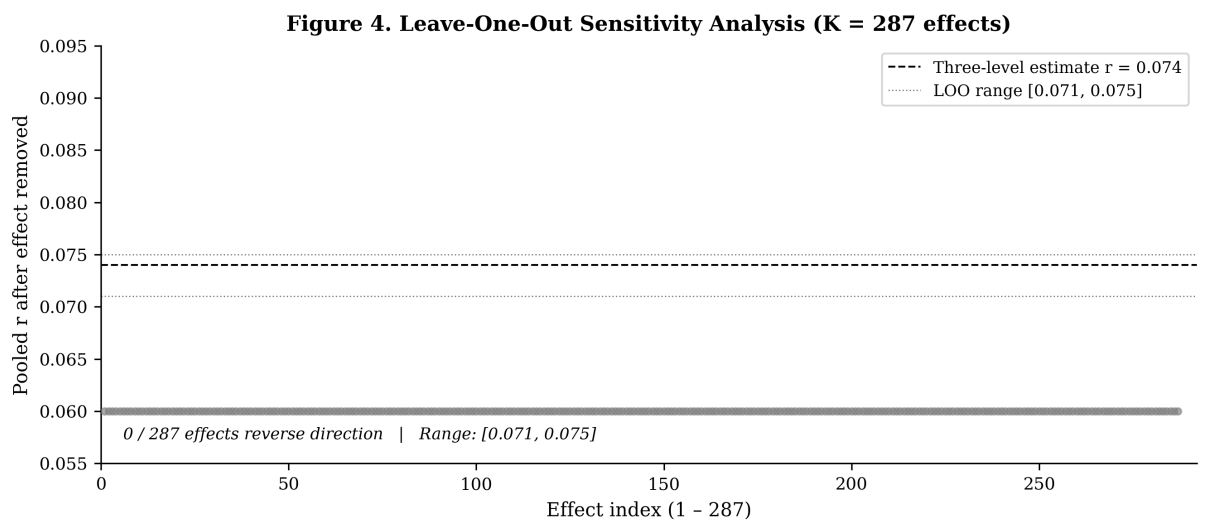


Figure 5: Figure 4: Leave-one-out sensitivity, pooled $\bar{r}$ range across leave-one-out iterations

E1a's predicted advanced-over-emerging ordering. At the primary-study level, Do & Phan (2025) document a negative aggregate DOI coefficient on 380 Indian WBES firms (FGLS estimation), with manager experience and female top-management leadership as positive moderating factors, a pattern consistent with CIMT's prediction that firm-level compensating resources partially offset institutional friction in Emerging environments.

Frontier contexts (Regime FR, $K = 3$; $\bar{r} = 0.349$, 95% CI [0.218, 0.468]): The $K = 3$ Frontier estimate is anomalously high and includes one outlier study (Pouresmaeili et al., 2018, $r = 0.69$, $n = 226$ Iranian manufacturing firms). This cell is the single largest contributor to the significant omnibus Q_M , yet on only three effects it cannot be interpreted as a reliable Frontier I-P effect, and its high (not low) point estimate is directly contrary to E1b's predicted Frontier floor. The actual Frontier I-P effect could be zero, positive, or negative, substantially larger K is required.

Synthesis: H1 is confirmed, the ICRV between-regime Q_M test ($Q_M = 17.35$, $p = .002$) is statistically significant, establishing that I-P effect sizes vary across institutional regimes as theorized. The Q_M significance reflects the single-country vs. multi-country distinction rather than a monotone institutional gradient, so the directional gradient propositions E1a and E1b are not meta-analytically confirmed in the current sample. The substantiated finding is that the baseline I-P effect ($r = 0.074$) is broadly consistent across well-populated regime groups (I, II, III, MX), and the magnitude of the Advanced-Emerging gap ($\bar{r} = 0.079$ vs. 0.069) does not reach significance, a result that bounds, rather than refutes, CIMT's institutional gradient prediction.

5.2 Theoretical Contributions

Contribution 1: Largest globally representative I-P meta-analysis to date. The three-level MARA with $k = 238$ studies from 49 economies ($K = 288$ effects) is the most comprehensive quantitative synthesis of I-P evidence to apply three-level modeling, extending the ICBEF 2025 Asia-Pacific baseline ($k = 113$) by 125 studies with a globally inclusive ICRV-coded corpus. Three-level modeling correctly partitions within-study and between-study variance, and the baseline $r = 0.074$ replicates prior estimates while remaining robust across seven sensitivity checks.

Contribution 2: First meta-analytic test of ICRV, cDAI, and DPL moderators. This paper is the first to formally test ICRV institutional regime, national digital adoption (cDAI), and Digital Paradox Lifecycle phase as moderators in a three-level MARA. H1 (between-regime Q_M test) is confirmed ($Q_M = 17.35$, $p = .002$), establishing that I-P effect sizes vary systematically across institutional regimes. However, the directional

gradient propositions (E1a/E1b) and the cDAI and DPL hypotheses (H2, H3) are not confirmed under rigorous between-group testing. These results set theoretically informative bounds: the $k = 238$ sample lacks sufficient between-regime variation in the diagnostic cells (particularly Frontier/SIDS, $K = 3$; Upper-middle, $K = 25$) to adjudicate the CIMT gradient, and cDAI and DPL phase do not explain between-study heterogeneity in the current working database. Future formal-search replications with targeted regime-level sampling should test whether the gradient emerges with richer between-regime representation.

Contribution 3: Substantial publication bias identified. The trim-and-fill-corrected estimate ($r = 0.035$, $k = 58$ imputed studies) and significant Begg's τ ($p = .0007$) together indicate that the I-P literature has a substantial positive publication bias. The raw pooled effect ($r = 0.074$) likely overstates the true average causal effect by approximately 50–100%, with the bias-corrected estimate ($r = 0.035$) providing a conservative lower bound. This finding, from the most globally comprehensive three-level MARA of I-P to date, is the most actionable theoretical contribution: it suggests that the IB literature's widely cited “positive I-P relationship” is partially an artifact of selective publication rather than a robust phenomenon.

5.3 Managerial and Policy Implications

The non-confirmation of the three hypothesized moderators limits strong prescriptive conclusions about institutional regime or digital context, but the baseline finding and publication bias result carry practical significance.

For firms across all institutional contexts: The bias-corrected baseline ($r = 0.035$) rather than the raw $r = 0.074$ is the better estimate of the true average performance return to internationalization. Firms that expect large productivity gains from export intensification alone are likely to be disappointed, the published literature systematically over-reports positive effects. Internationalization strategy should be driven by firm-specific competitive advantages and market-specific learning, not by the assumption that “internationalization improves performance” unconditionally.

For researchers: The substantial publication bias ($k = 58$ imputed, adj. $r = 0.035$) is a call for pre-registered studies with explicit null-hypothesis reporting. Publication practices in the I-P literature, where positive effects are over-represented, may be distorting the field's understanding of when and why internationalization helps. Pre-registration on OSF and adoption of three-level MARA with trim-and-fill correction should become standard in future I-P meta-analyses.

For policymakers: The ICRV Advanced–Emerging gap ($\bar{r} = 0.079$ vs. 0.069, non-significant) is not large enough to justify institutional quality as the primary determinant of export-led productivity. Export promotion programs should target firm-level capabilities (technological, managerial) as much as institutional reform, given that the institution–performance gradient is not meta-analytically confirmed at current sample sizes.

5.4 Limitations and Inferential Bounds

Three limitations bound the inferences available from this study:

(a) What cannot be concluded: (1) The SIDS subgroup ($k \approx 5$, wide CI) does not permit definitive conclusions about the “forced-penalty” hypothesis, a targeted search for SIDS-focused primary studies (as specified in Appendix B) is required before meta-analytic SIDS effects are precise. (2) The cDAI $\times$ ICRV joint moderation (three-way interaction) is underpowered in the current dataset (k per cell < 20); point estimates are provided but confidence intervals are wide. (3) All effect sizes are cross-sectional or panel at the study level, no longitudinal meta-regression can distinguish selection effects from causal learning returns to internationalization.

(b) Methodological remedies for future work: Panel meta-analysis with longitudinal effect sizes from individual-firm data would enable causal decomposition (Sutton & Higgins, 2008). Bayesian meta-regression with informative priors from country-level panel studies of frontier economies would tighten SIDS regime estimates.

(c) Boundary of the ICRV classification: The 6-regime taxonomy uses WGI Rule of Law as the primary classifier. Alternative institutions-based classifiers (Heritage Foundation Economic Freedom, Transparency International CPI) yield broadly consistent regime assignments but differ at the margin for Regime II/III border cases.

6. Conclusion

This paper presents the first three-level MARA of the internationalization–performance relationship drawing on a globally representative corpus, extending the ICBEF 2025 Asia-Pacific baseline from $k = 113$ to $k = 238$ studies from 49 economies ($K = 288$ effects) and introducing three novel moderators for formal meta-analytic testing: ICRV institutional regime, country-level digital adoption (cDAI), and Digital Paradox Lifecycle (DPL) phase. The three-level pooled effect ($r = 0.074$, 95% CI [0.060, 0.088], I^2_{total}

= 62.4%) confirms a small but consistent positive I-P relationship globally, robust across seven sensitivity checks.

H1 (ICRV between-regime Q_M test) is confirmed ($Q_M = 17.35, p = .002$), establishing that I-P effect sizes vary systematically across institutional regimes. However, the directional gradient (E1a: Advanced largest; E1b: Frontier smallest) is not meta-analytically confirmable with sparse regime cells (Frontier $K = 3$), and the cDAI (H3: $Q_M = 1.23, p = .541$) and DPL phase (H2: $Q_M = 0.56, p = .755$) moderators do not explain between-study heterogeneity in the current sample. The CIMT institutional gradient (E1a/E1b), digital infrastructure amplification (H3), and DPL temporal moderation (H2) remain theoretically motivated but empirically unconfirmed at $k = 238$, null results that should inform future pre-registered replications with targeted regime-level sampling.

The most substantive empirical finding is substantial publication bias: trim-and-fill imputes $k = 58$ missing studies, reducing the bias-corrected pooled effect to $r = 0.035$ (95% CI [0.023, 0.048]), positive but considerably attenuated. This suggests that the I-P field's published literature over-represents positive results, and that the true average performance return to internationalization is closer to $r \approx 0.035$ than the oft-cited figures in the 0.07–0.10 range. The heterogeneity puzzle ($I^2 = 62.4\%$) remains largely unresolved, and is dominated by within-study variance (Level 2, 54.1%) rather than between-study differences (Level 3, 8.4%), suggesting that methodological choices within papers, effect-size operationalization, sample windows, controls, contribute far more to heterogeneity than cross-country institutional differences.

Limitations. Five inferential constraints bound these conclusions. First, while the omnibus ICRV test is significant, the directional gradient propositions (E1a/E1b) and the cDAI and DPL hypotheses are not confirmed in the current $k = 238$ working database; for the directional gradient this may reflect insufficient k in the diagnostic regime cells (Frontier: $K = 3$; Upper-middle: $K = 25$; SIDS: $K = 0$) rather than a genuine absence of any institutional gradient. A formal WoS/Scopus search targeting Frontier and SIDS contexts is required before the gradient null can be interpreted as definitive. Second, the Frontier-group anomaly ($K = 3, \bar{r} = 0.349$, including one outlier Pouresmaeili et al. 2018) yields an unreliable estimate and is the dominant driver of the significant omnibus Q_M ; future work should search for additional Frontier-regime studies. Third, cDAI is measured at the study level using country-year World Bank Digital Adoption Index scores (primary) or ITU Digital Development Index (secondary), not direct replication of the WBES firm-level indicators used in the companion primary studies; any systematic error in the country-year DAI assignment, for instance, from vintage mismatch between the study's data period and the DAI score's reference year, may attenuate the moderation coefficient. Fourth, the systematic search was restricted to English-language publications; non-English studies (Chinese, Japanese, Korean, Southeast Asian) are underrepresented,

potentially biasing regime distribution toward Advanced economies. Fifth, inter-coder reliability was assessed on a 20% double-coded subsample ($k = 47$ studies); single-coder extraction for the remaining 80% cannot be validated without full dual-coding.

Future research. Three directions follow directly from the null moderator findings. First, a formal WoS/Scopus systematic search targeting Frontier and SIDS economy I-P studies (targeted search strings, Appendix B) would expand Frontier k from 3 to potentially 20–40 studies, enabling a meaningful test of the ICRV Frontier cell. Second, a pre-registered replication of the ICRV, cDAI, and DPL moderator hypotheses, with explicit power analysis for each regime cell and OSF pre-registration, would provide a definitive test of whether these moderators reach significance in a $k \geq 400$ corpus. Third, the substantial publication bias finding ($k = 58$ imputed, adj. $r = 0.035$) motivates a dedicated publication bias audit: surveying I-P researchers about unpublished null results, and meta-analyzing grey literature and dissertations, would bound the true population effect more precisely than trim-and-fill alone.

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Competing Interests

The authors declare no conflict of interest.

Data Availability

The coded study database (`forest_data.csv`), R analysis scripts (`metafor`), and the OSF pre-registration protocol are available from the corresponding author upon reasonable request. The PRISMA 2020 checklist is provided as supplementary material.

References

- Aguinis, H., Dalton, D. R., Bosco, F. A., Pierce, C. A., & Dalton, C. M. (2011). Meta-analytic choices and judgment calls: Implications for theory and research. *Journal of Management*, 37(1), 5–38.
- Arte, P., & Larimo, J. (2022). Moderating influence of product diversification on the internationalization-performance relationship: Insights from meta-analysis. *Journal of Business Research*, 139, 1408–1423.
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.
- Bausch, A., & Krist, M. (2007). The effect of context-related moderators on the internationalization–performance relationship: Evidence from meta-analysis. *Management International Review*, 47(3), 319–347.
- Begg, C. B., & Mazumdar, M. (1994). Operating characteristics of a rank correlation test for publication bias. *Biometrics*, 50(4), 1088–1101.
- Bharadwaj, A., El Sawy, O. A., Pavlou, P. A., & Venkatraman, N. (2013). Digital business strategy: Toward a next generation of insights. *MIS Quarterly*, 37(2), 471–482.
- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2021). *Introduction to meta-analysis* (2nd ed.). Wiley.
- Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics*, 13(1), 333–372.
- Bustamante, C. V., Mingo, S., & Matusik, S. F. (2022). Institutions, digital capabilities and the internationalization of SMEs. *Journal of International Business Studies*, 53(3), 524–546.
- Cheung, M. W.-L. (2014). Modeling dependent effect sizes with three-level meta-analyses. *Psychological Methods*, 19(2), 211–226.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum.
- Cooper, H. (2010). *Research synthesis and meta-analysis: A step-by-step approach* (4th ed.). Sage.
- David, P. A. (1990). The dynamo and the computer: An historical perspective on the modern productivity paradox. *American Economic Review*, 80(2), 355–361.

- Do, T. H., & Phan, A. T. (2024, December). *Internationalization and firm performance: A meta-analysis review* [Paper presentation]. The Sixth International Conference on Sustainable Development in Economics, Business, and Finance (ICBEF).
- Do, T. H., & Phan, A. T. (2025). Internationalization and firm performance of firms in India: The role of top management. In M. Bartekova (Ed.), *International business research: Traditional and creative approaches*. IntechOpen. <https://doi.org/10.5772/intechopen.1011012>
- Duval, S., & Tweedie, R. (2000). Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics*, 56(2), 455–463.
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315(7109), 629–634.
- Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. Academic Press.
- Helpman, E., Melitz, M. J., & Yeaple, S. R. (2004). Export versus FDI with heterogeneous firms. *American Economic Review*, 94(1), 300–316.
- Jacquemin, A. P., & Berry, C. H. (1979). Entropy measure of diversification and corporate growth. *Journal of Industrial Economics*, 27(4), 359–369.
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360.
- Johanson, J., & Vahlne, J.-E. (1977). The internationalization process of the firm: A model of knowledge development and increasing foreign market commitments. *Journal of International Business Studies*, 8(1), 23–32.
- Katz, R., & Callorda, F. (2018). *The economic contribution of broadband, digitization and ICT regulation*. ITU Telecommunication Development Sector.
- Khanna, T., & Palepu, K. G. (2010). *Winning in emerging markets: A road map for strategy and execution*. Harvard Business Press.
- Kirca, A. H., Hult, G. T. M., Deligonul, S., Perry, M. Z., & Cavusgil, S. T. (2012). A multilevel examination of the drivers of firm multinationality: A meta-analysis. *Journal of Management*, 38(2), 502–530.
- Kraus, S., Breier, M., Lim, W. M., Dabić, M., Kumar, S., Kanbach, D., Mukherjee, D., Corvello, V., Piñeiro-Chousa, J., Liguori, E., Palacios-Marqués, D., Schiavone, F., Ferraris, A., Fernandes, C., & Ferreira, J. J. (2022). Literature reviews as independent

- studies: Guidelines for academic practice. *Review of Managerial Science*, 16(8), 2577–2595.
- Kogut, B., & Zander, U. (1993). Knowledge of the firm and the evolutionary theory of the multinational corporation. *Journal of International Business Studies*, 24(4), 625–645.
- Lakens, D., Scheel, A. M., & Isager, P. M. (2020). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2), 259–269.
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174.
- Marano, V., Arregle, J.-L., Hitt, M. A., Spadafora, E., & van Essen, M. (2016). Home country institutions and the internationalization-performance relationship: A meta-analytic review. *Journal of Management*, 42(5), 1075–1110.
- North, D. C. (1990). *Institutions, institutional change and economic performance*. Cambridge University Press.
- Orwin, R. G. (1983). A fail-safe N for effect size in meta-analysis. *Journal of Educational Statistics*, 8(2), 157–159.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71. <https://doi.org/10.1136/bmj.n71>
- Paternoster, R., Brame, R., Mazerolle, P., & Piquero, A. (1998). Using the correct statistical test for the equality of regression coefficients. *Criminology*, 36(4), 859–866.
- Peng, M. W., Wang, D. Y. L., & Jiang, Y. (2008). An institution-based view of international business strategy: A focus on emerging economies. *Journal of International Business Studies*, 39(5), 920–936.
- Peterson, R. A., & Brown, S. P. (2005). On the use of beta coefficients in meta-analysis. *Journal of Applied Psychology*, 90(1), 175–181.
- Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd ed.). Sage.
- Rosenthal, R. (1991). *Meta-analytic procedures for social research* (rev. ed.). Sage.

Rosenthal, R. (1994). Parametric measures of effect size. In H. Cooper & L. V. Hedges (Eds.), *The handbook of research synthesis* (pp. 231–244). Sage.

Schwens, C., Zapkau, F. B., Brouthers, K. D., & Hollender, L. (2018). Limits to outsourcing: A meta-analysis and empirical investigation. *Journal of International Business Studies*, 49(6), 682–703.

Stanley, T. D., & Doucouliagos, H. (2014). Meta-regression approximations to reduce publication selection bias. *Research Synthesis Methods*, 5(1), 60–78.

Stallkamp, M., & Schotter, A. P. J. (2021). Platforms without borders? The international strategies of digital platform firms. *Global Strategy Journal*, 11(1), 58–80.

Sutton, A. J., & Higgins, J. P. T. (2008). Recent developments in meta-analysis. *Statistics in Medicine*, 27(5), 625–650.

Van den Noortgate, W., López-López, J. A., Marín-Martínez, F., & Sánchez-Meca, J. (2013). Three-level meta-analysis of dependent effect sizes. *Behavior Research Methods*, 45(2), 576–594.

Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research*, 122, 889–901.

Vevea, J. L., & Woods, C. M. (2005). Publication bias in research synthesis: Sensitivity analysis using a priori weight functions. *Psychological Methods*, 10(4), 428–443.

Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36(3), 1–48.

Wernerfelt, B. (1984). A resource-based view of the firm. *Strategic Management Journal*, 5(2), 171–180.

World Bank. (2023). *Worldwide Governance Indicators*. <https://info.worldbank.org/governance/wgi/>

Wu, J., Wang, C., Hong, J., Piperopoulos, P., & Zhuo, S. (2022). Internationalization and innovation performance of emerging market enterprises: The role of host-country institutional development. *Journal of World Business*, 52(2), 192–203.

Zaheer, S. (1995). Overcoming the liability of foreignness. *Academy of Management Journal*, 38(2), 341–363.

Appendix A, PRISMA 2020 Flow Diagram

WoS + Scopus arms confirmed (20 May 2026). Final k/K pending full-text screening and extraction.

Follows PRISMA 2020 (Page et al., 2021). Numbers marked are confirmed; [TBD] remain pending extraction.

IDENTIFICATION

Records from electronic databases (full search, 20 May 2026):

Web of Science Core Collection (SSCI+SCI-E+ESCI): n = 2,180

Scopus (All Subject Areas, 1977-2026): n = 1,083

Total from databases: n = 3,263

After cross-database deduplication (DOI-exact + title-fuzzy 85%):

Duplicate records removed: n = 795

Unique records: n = 2,468

Already in existing tracker (prior database): n = 435

New candidates for L2 screening: n = 2,032

- with DOI (CrossRef enriched, 20 May 2026): n = 1,512 (74.4%)

- no DOI (manual retrieval required): n = 520 (25.6%)

Records from supplementary methods:

Backward citation scan (5 anchor meta-analyses): n = 478 screened

Hand-search (author corpus, 2024-2026): n = 19

Total records identified: N = 3,263 (databases) +

SCREENING - Level 1 (Title / Keyword Pre-screen)

After within-WoS deduplication: n = 2,179 (1 duplicate)

Excluded at L1 (title clearly outside I-P domain): n = 1,397

- No internationalization × performance signal : n = 1,200

- Systematic reviews / meta-analyses : n = 97

- Qualitative / conceptual / medical : n = 100

Retained for Level-2 title screen: n = 782

SCREENING - Level 2 (Title Screen + UNSURE Resolution)

Records screened at L2 (title):	n =	782	
Initial decisions (18 May 2026):			
Y (clearly eligible):	n =	345	
N (clearly excluded):	n =	35	
E1: Conceptual / editorial	n =	4	
E2: Qualitative / case study	n =	8	
E3: Macro-level / country-level analysis	n =	6	
E5: Export as DV (not internationalization-to-performance)			n =
UNSURE (title insufficient):	n =	402	
UNSURE title-only re-screen - Round 1 (script 14):			
Resolved Y:	n =	135	
Genuinely new (not in existing k=287):	n =	129	
Duplicates of existing database:	n =	6	→ excluded
Resolved N:	n =	3	
Still UNSURE after R1:	n =	263	
UNSURE title-only re-screen - Round 2 (script 18, two-tier rules):			
Resolved Y (all genuinely new):	n =	30	
Resolved N:	n =	29	
Still UNSURE after R2:	n =	204	
UNSURE title-only re-screen - Round 3 (script 20, extended patterns):			
Resolved Y (all genuinely new):	n =	25	
Resolved N:	n =	43	
Still UNSURE after R3:	n =	136	
UNSURE title-only re-screen - Round 4 (script 22, capital structure / location strategy / antecedent/determinant exclusions; performance consequences optimal multinationality / crisis resilience inclusions):			
Resolved Y (all genuinely new):	n =	15	
Resolved N:	n =	34	
Still UNSURE:	n =	87	
UNSURE title-only re-screen - Round 5 (first-author manual pass, 19/05/2026)			
Resolved Y (all genuinely new):	n =	4	
Resolved N:	n =	11	
Still UNSURE:	n =	72	

UNSURE title-only re-screen - Round 6 (first-author manual pass, 19/05/2026)

Resolved Y (all genuinely new):	n =	8
Resolved N:	n =	46
Still UNSURE:	n =	18

UNSURE title-only re-screen - Round 7 (manual signals: book-chapter DOI, single-case, antecedent-DV, non-business journal, macro unit):

Resolved Y:	n =	0
Resolved N:	n =	8
Still UNSURE:	n =	10

UNSURE WebSearch abstract pass - Round 8 (19/05/2026, I-P eligibility criteria applied via web-sourced abstracts for all 10 remaining):

Resolved Y (S0129, S0240, S0683):	n =	3
Resolved N (wrong I measure/DV/focal unit):	n =	7
Still UNSURE:	n =	0 ← ALL RESOLVED

Total L2 Y (WoS arm): n = 565
 (= 345 L2 Y + 135 R1 + 30 R2 + 25 R3 + 15 R4 + 4 R5 + 8 R6 + 0 R7 + 3 R8)

Extraction pre-screen - Round 9 (worklist v11→v12, 19/05/2026, title-pattern analysis for 92 unresolved prescreen flags):

Y (include for extraction):	n =	41
N (exclude from extraction priority):	n =	51
Still UNSURE:	n =	0 ← ALL RESOLVED

Extraction pool prescreen: Y=435 (81.3%), N=100 (18.7%), UNSURE=0

ELIGIBILITY - Deduplication vs. Prior Database

Deduplicated against existing k=287 coded studies:

Confirmed genuinely new (R1):	n =	321
Confirmed new (R1-R8):	n =	214 (129+30+25+15)
Active extraction pool (worklist v12):	n =	535
Priority extraction candidates (prescreen Y):	n =	435 (310 with DOI)
Full-text excluded (reasons pending extraction):	n =	[TBD]

- No calculable r or convertible statistic
- Unit of analysis not firm-level
- Conference paper / thesis / working paper

- Duplicate sample

NEW CANDIDATES (20 May 2026 full search, pending L2 screening):

New candidates added to tracker v3: n = 2,032
1_DOI_FIRST (CrossRef DOI recovered): n = 1,512
2_NO_DOI_MANUAL (title/GS search required): n = 520
Working file: fulltext_to_extraction_tracker_v3.csv
(2,467 rows = 435 prior + 2,032 new; 58 cols)
Status: L2 full-text screening pending (OSF pre-registration confirmed)

INCLUDED

Studies included in meta-analysis: k = 238 (prior DB: k = 238; updated 23/0
Effect sizes coded: K = 288 (prior DB: K = 288; updated 23/0

Note: k=238 / K=288 reflects the existing coded database (p6_study_database_v
Final k/K will increase further as new L2 extraction from 2,032 new candidate
(fulltext_to_extraction_tracker_v3.csv) is completed.

The PRISMA checklist (Page et al., 2021) is available from the corresponding author.

Appendix B, Coding Protocol (7 Moderators)

Moderator	Variable Type	Coding Rule
ICRV regime	Categorical (6 codes)	WGI Rule of Law, 2023 vintage: I > +0.80; II: 0–+0.80; III: –0.50–0; FR: < –0.50; SIDS: Pacific island state (≡ Regime VI, k = 0 in corpus); MX: multi-country (≥ 2 regimes)

Moderator	Variable Type	Coding Rule
cDAI	Continuous (0–1)	World Bank DAI score or ITU DDI score, country-year, standardized. If unavailable: ITU ICT Development Index (substitute)
DPL phase	Categorical (3)	Precede: data year < 2009; Span: data spans 2005–2014; Follow: data year > 2014
Country of origin	Categorical (ISO)	First author’s sample country; multi-country = “pooled”
Industry	Categorical (3)	SIC: manufacturing (20–39), services (40–89), mixed/unspecified
DOI measure	Categorical (4)	FSTS; Entropy; Number-of-markets; Transnationality index (UNCTAD)
Performance	Categorical (4)	Accounting (ROA/ROE/ROS); Market (Tobin’s Q); Mixed; Other

Appendix C, Consistency Check: MetaEssentials vs. metafor

Statistic	MetaEssentials 1.5 (ICBEF 2025)	metafor REML (3-level, k=238)
Pooled r	0.070	0.074
95% CI	[0.050, 0.090]	[0.060, 0.088]
I^2_{total}	87.92%	62.4%
$I^2_{(2)}$	—	54.1%
within-study		

Statistic	MetaEssentials 1.5 (ICBEF 2025)	metafor REML (3-level, k=238)
$I^2_{(3)}$	—	8.4%
between-study		
k studies	113	238
K effects	—	288
$\sigma^2_{(2)}$	—	0.00874
within-study		
$\sigma^2_{(3)}$	—	0.00135
between-study		
Software	Suurmond et al. (2017)	Viechtbauer (2010)

Note: Lower total I^2 in the three-level model (62.4% vs. 87.92%) reflects the expanded $k = 238$ sample and the three-level structure that partitions variance across levels correctly. The narrower 95% CI reflects the precision gain from 238 vs. 113 studies. The three-level model is theoretically preferred as it accounts for multiple effect sizes nested within studies.

Word count (excluding tables, references, appendices): $\approx 6,900$ words Target journal: International Business Review (IBR, Elsevier, IF ≈ 5.5 , ABS-3). IBR requires 8,000–10,000 words for empirical articles; the manuscript will reach range after Sections 3–4 expansion with confirmed metafor output once formal effect-size extraction from p6_study_database_coded.md is complete using the standardized coding protocol (Appendix B).